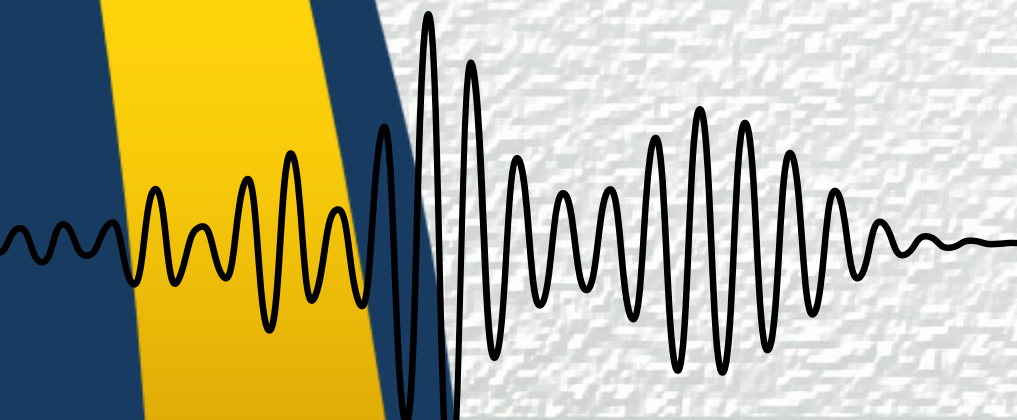
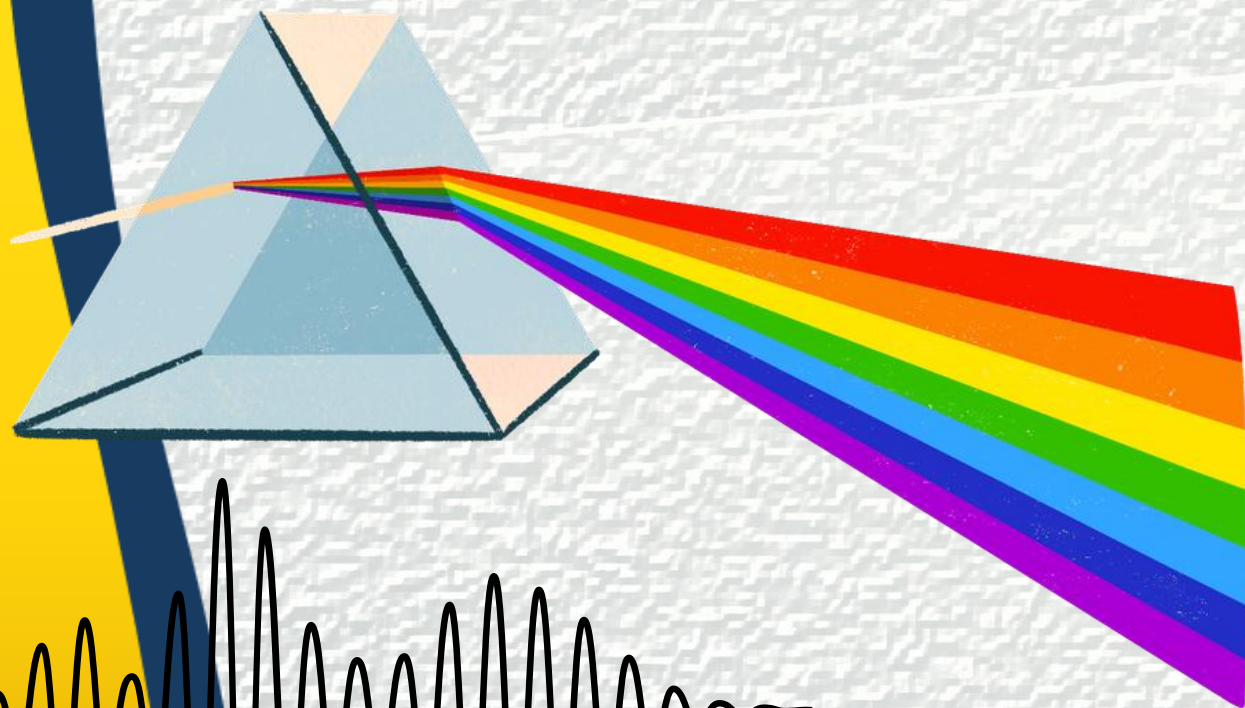


GENERAL SCIENCE 11

WEEK 7

UNDERSTANDING AND USING

LIGHT *and* **SOUND**



LEARNING OBJECTIVES:

At the end of the lesson, I should be able to:

- **Differentiate between a mechanical wave (sound) that requires a physical medium to travel and an electromagnetic wave (light) that can travel through a vacuum.**
- **Explain the 9 primary properties of light waves, including speed, reflection, refraction, and polarization.**





MECHANICAL VS. ELECTROMAGNETIC

- **Sound is a Mechanical Wave:** It is stubborn. It requires a medium (matter like air, water, or solids) to travel. Sound moves by vibrating particles and passing that energy to the next particle. No particles = no sound
- **Light is an Electromagnetic Wave:** It is a solo traveler. It consists of oscillating electric and magnetic fields and does not need a medium. It can travel flawlessly through the empty vacuum of space.



◀◀ INTRODUCTION TO LIGHT WAVES

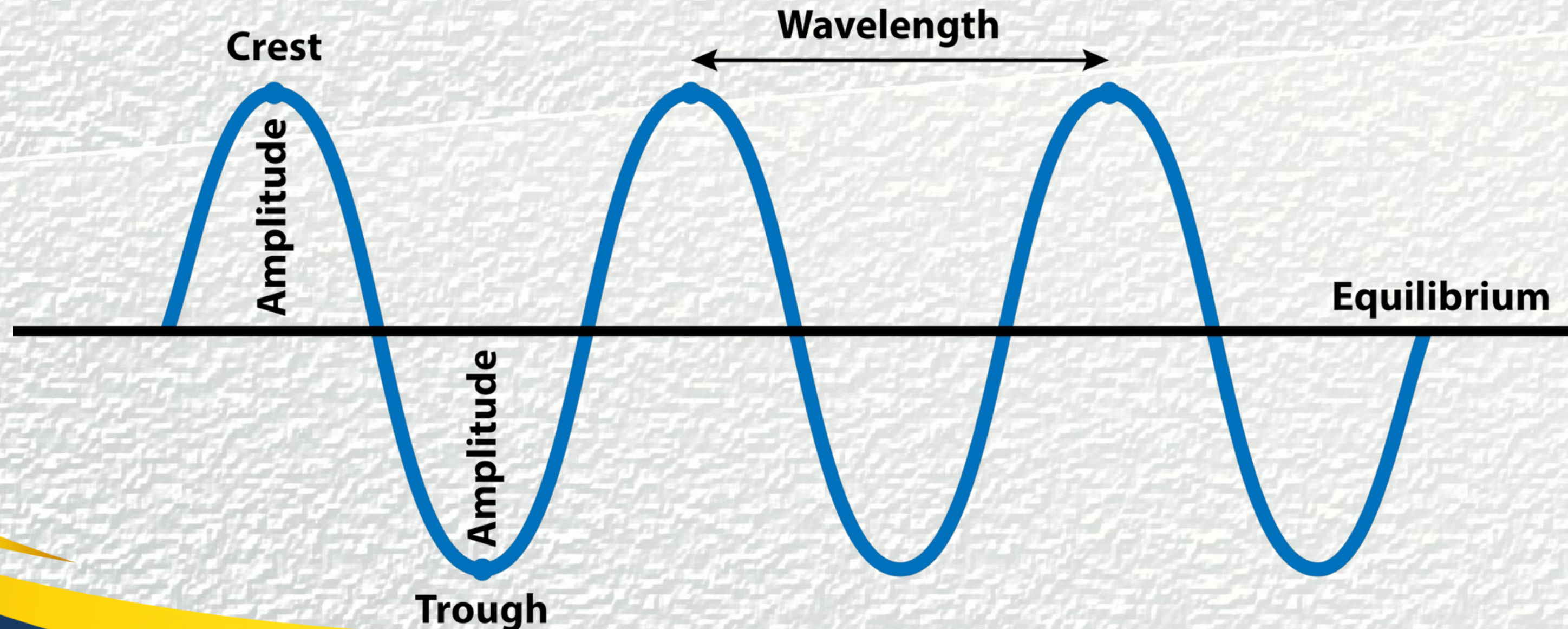
- **Light** is an electromagnetic wave that travels as a transverse wave (oscillating up and down, perpendicular to the direction of travel).
- Unlike sound, light does not need a medium. It can travel through the absolute empty vacuum of space.



◀◀ INTRODUCTION TO LIGHT WAVES

Parts of the Wave:

- **Crest:** The highest peak of the wave.
- **Trough:** The lowest dip of the wave.



« PROPERTIES OF LIGHT WAVES

1) SPEED

The Cosmic Speed Limit: In a perfect vacuum, light travels at its ultimate speed, denoted by the letter c:

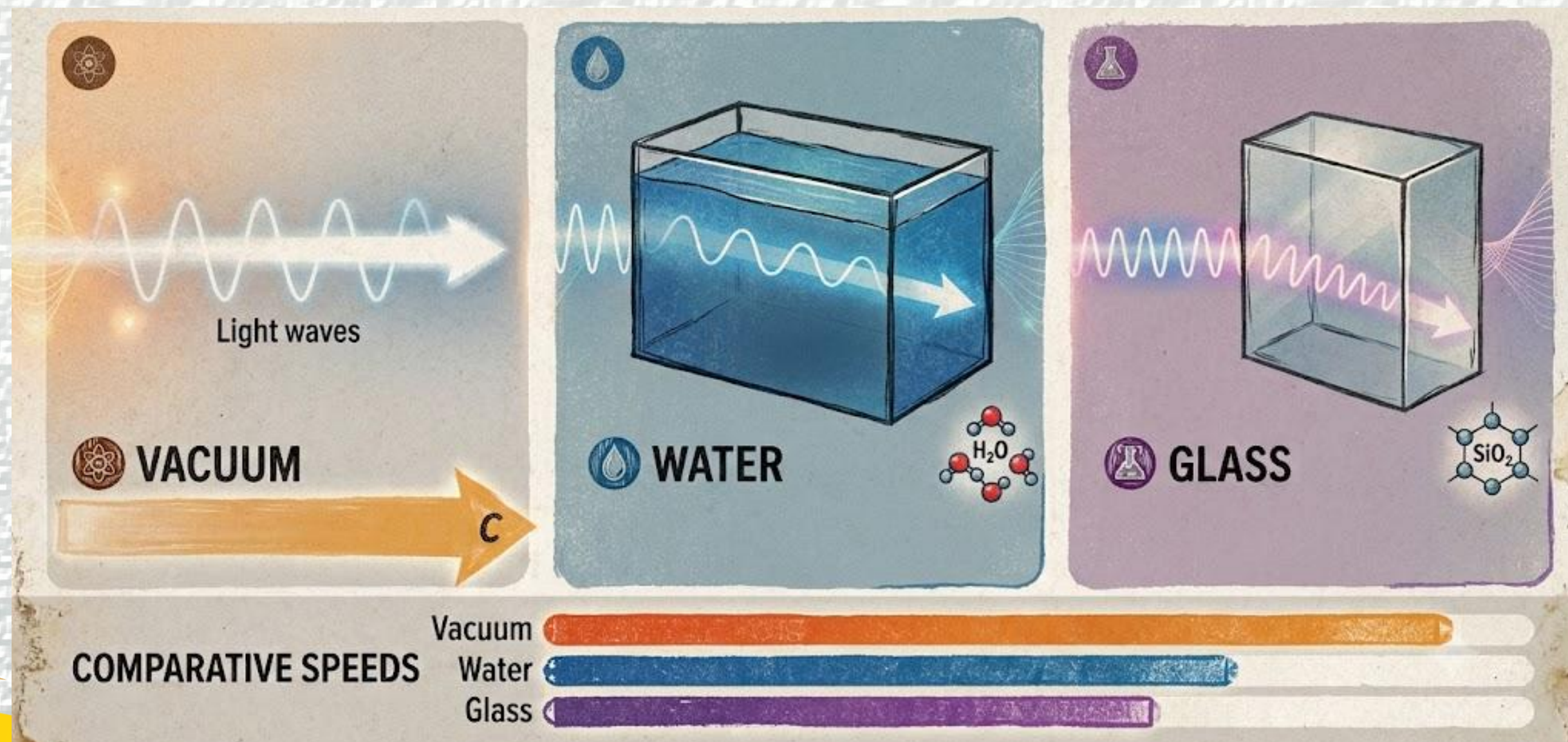
$$c = 300,000,000 \text{ m/s}$$
$$(3 \times 10^8 \text{ m/s})$$



« PROPERTIES OF LIGHT WAVES

SPEED

- Sound travels faster in dense materials, but light slows down when it passes through matter (like water or glass) because the particles get in the way of the electromagnetic fields.

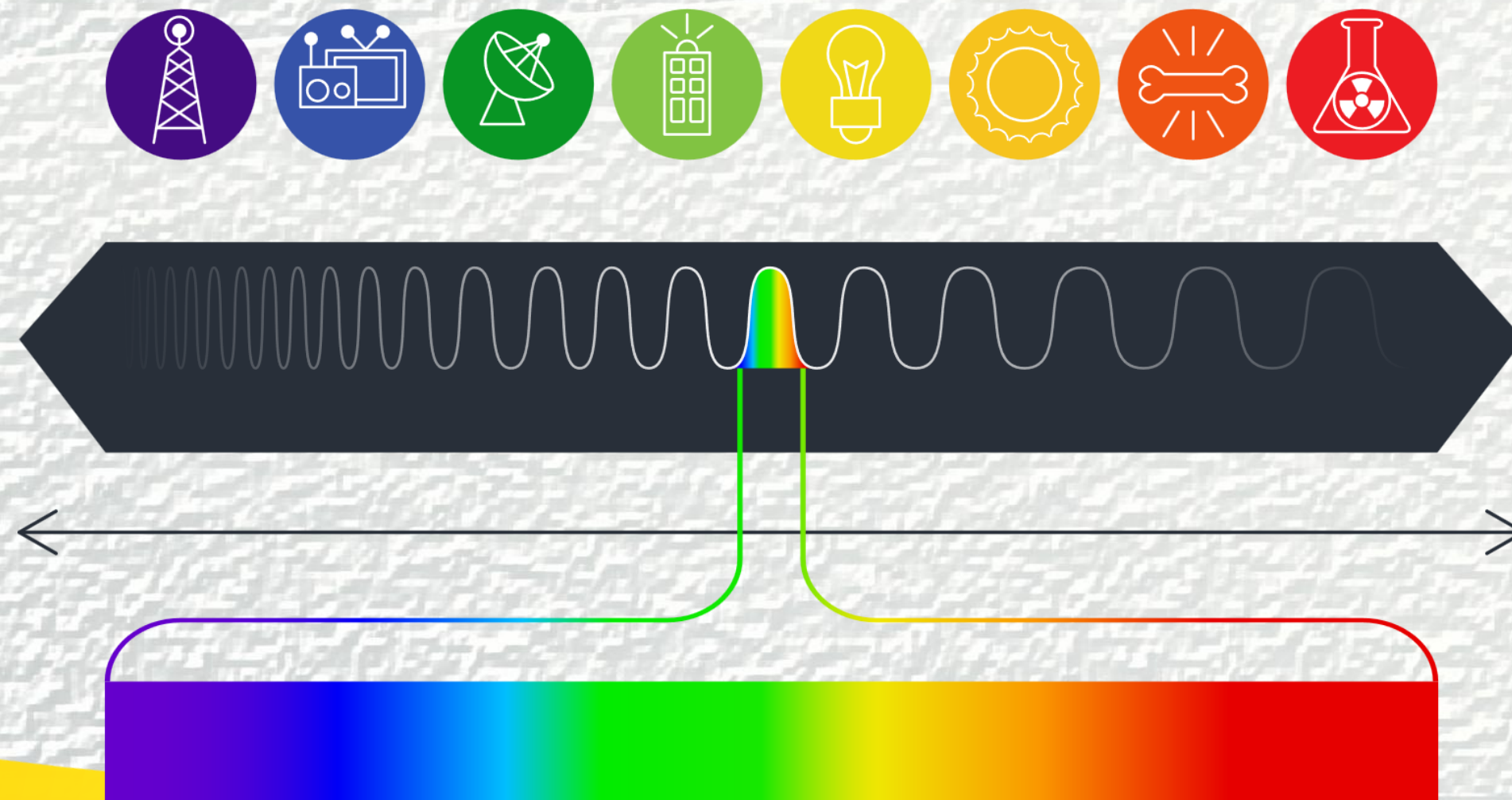


« PROPERTIES OF LIGHT WAVES

2-3) WAVELENGTH and FREQUENCY

Frequency (f): The number of wave crests that pass a point in one second. Measured in Hertz (Hz).

Wavelength: The physical distance from one crest to the very next crest. Measured in nanometers (nm) for visible light.

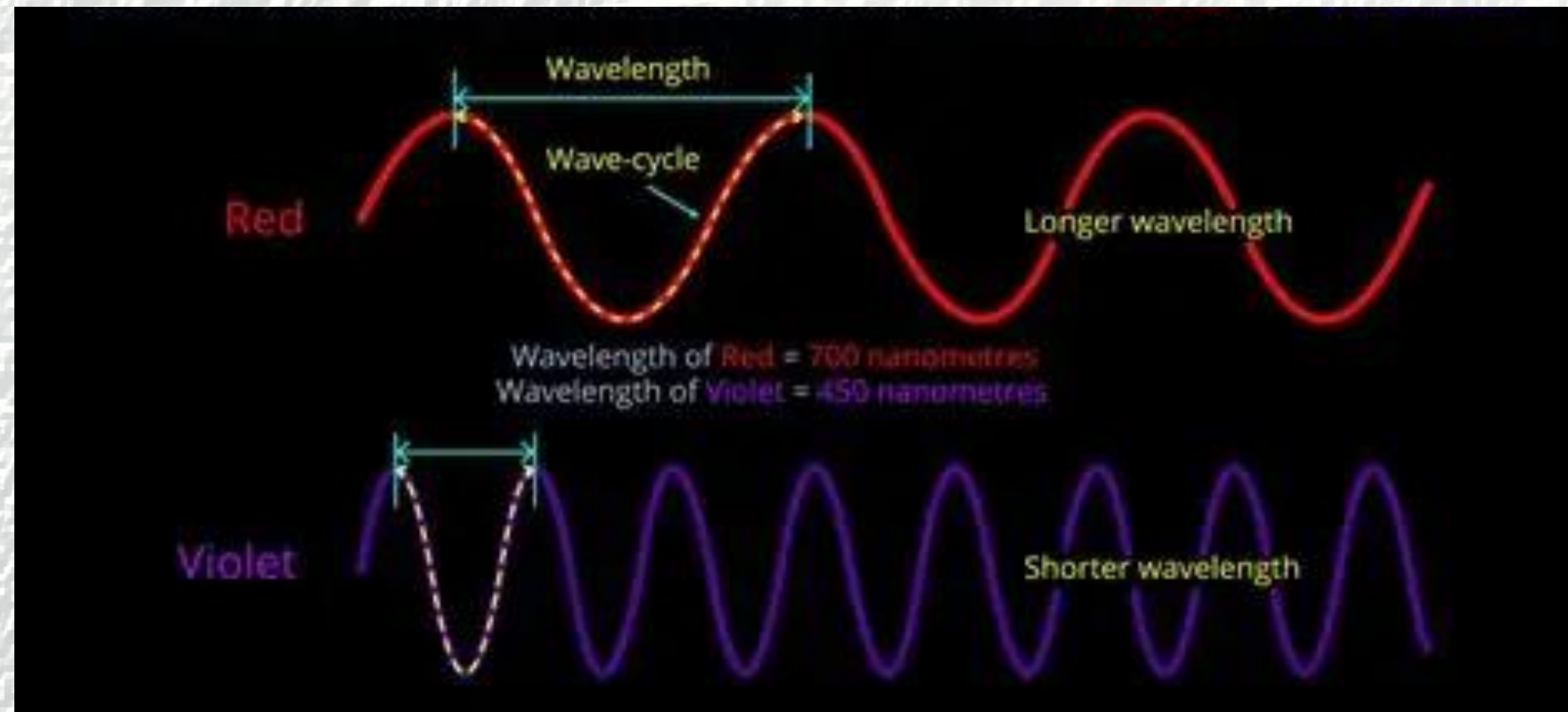


« PROPERTIES OF LIGHT WAVES

WAVELENGTH and FREQUENCY

Frequency and Wavelength dictate the color of light we see.

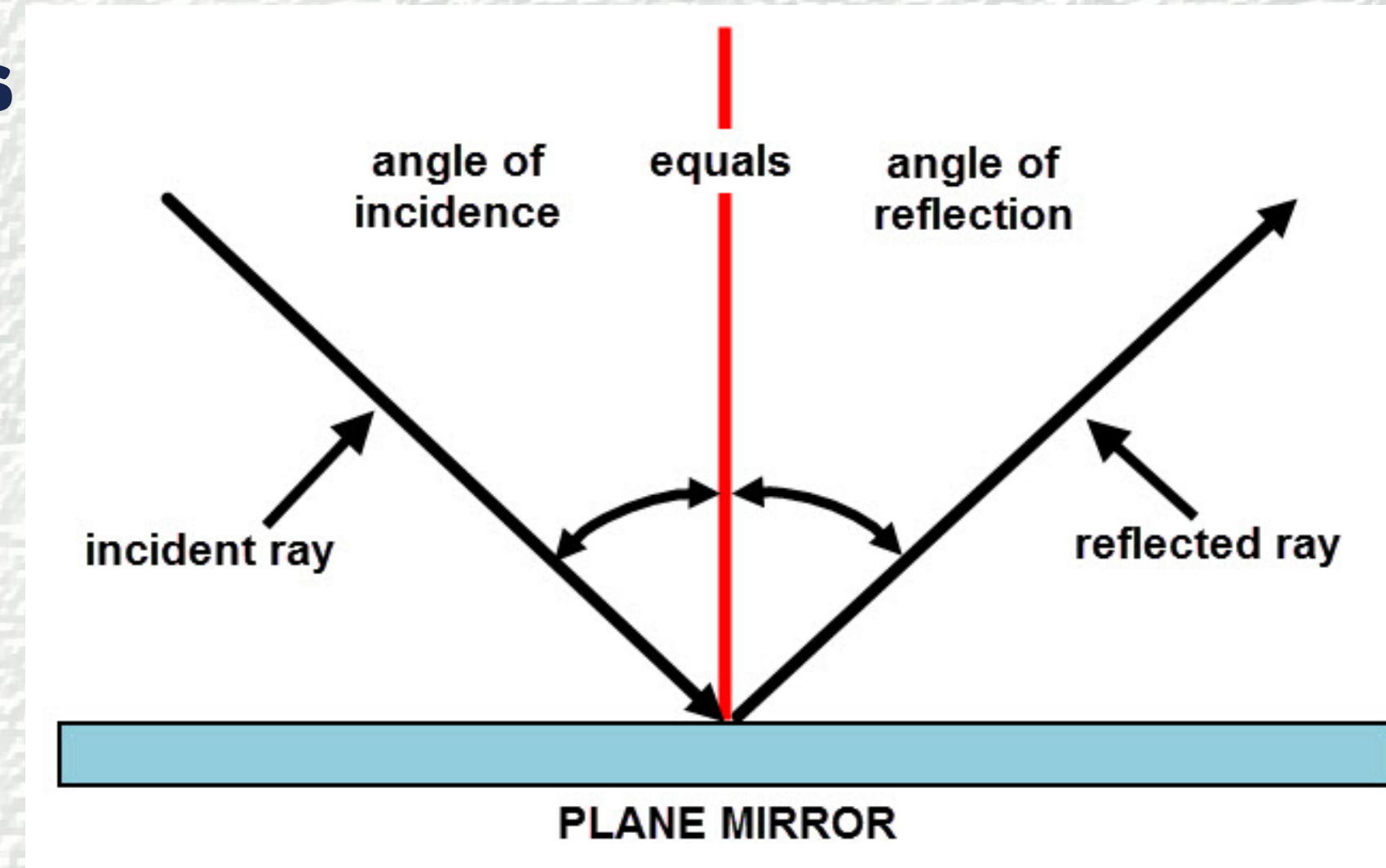
- **Red Light:** Has the longest wavelength and the lowest frequency.
- **Violet/Blue Light:** Has the shortest wavelength and the highest frequency.



« PROPERTIES OF LIGHT WAVES

4) REFLECTION

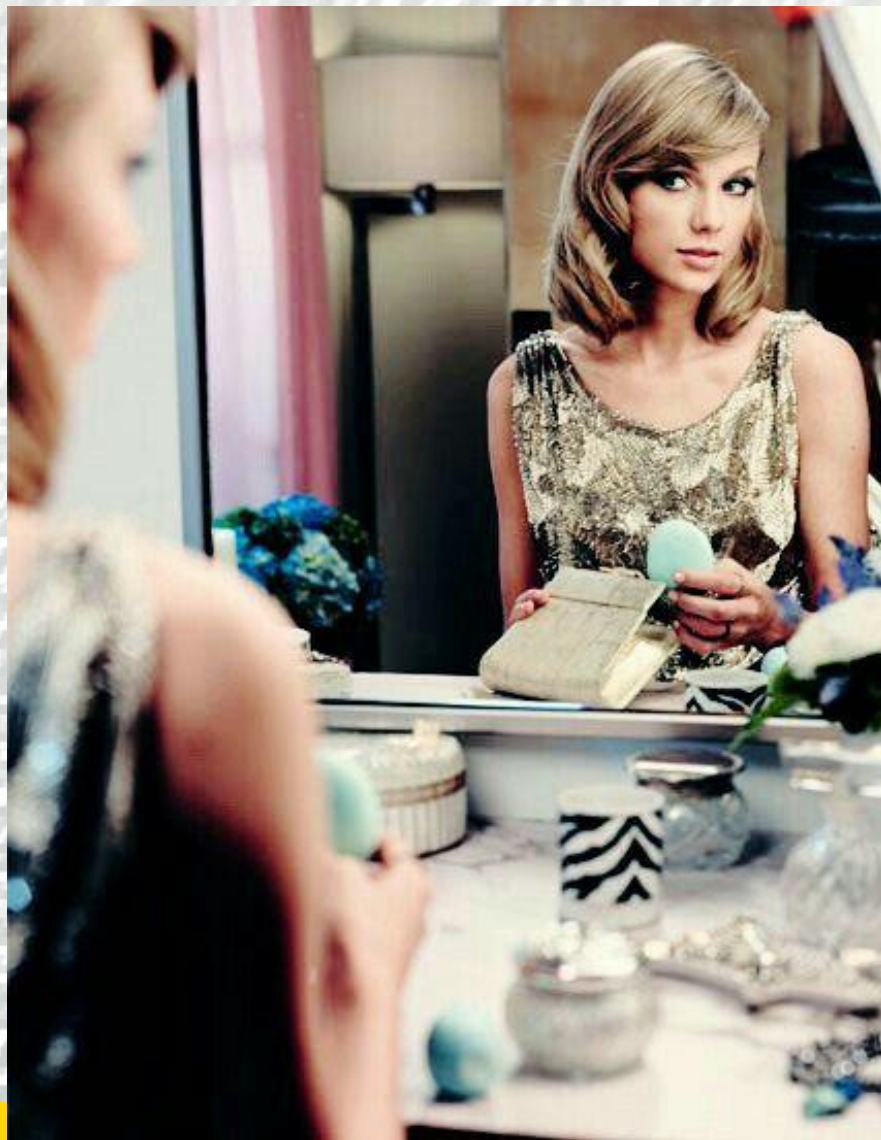
- Reflection happens when light waves hit a surface and bounce back into the original medium.
- The Law of Reflection: The angle at which the light hits the surface (Angle of Incidence) is exactly equal to the angle at which it bounces off (Angle of Reflection).



« PROPERTIES OF LIGHT WAVES

REFLECTION

- **Real-World Example: Looking into a smooth mirror, or seeing the moon reflect off a calm lake.**



« PROPERTIES OF LIGHT WAVES

5) REFRACTION

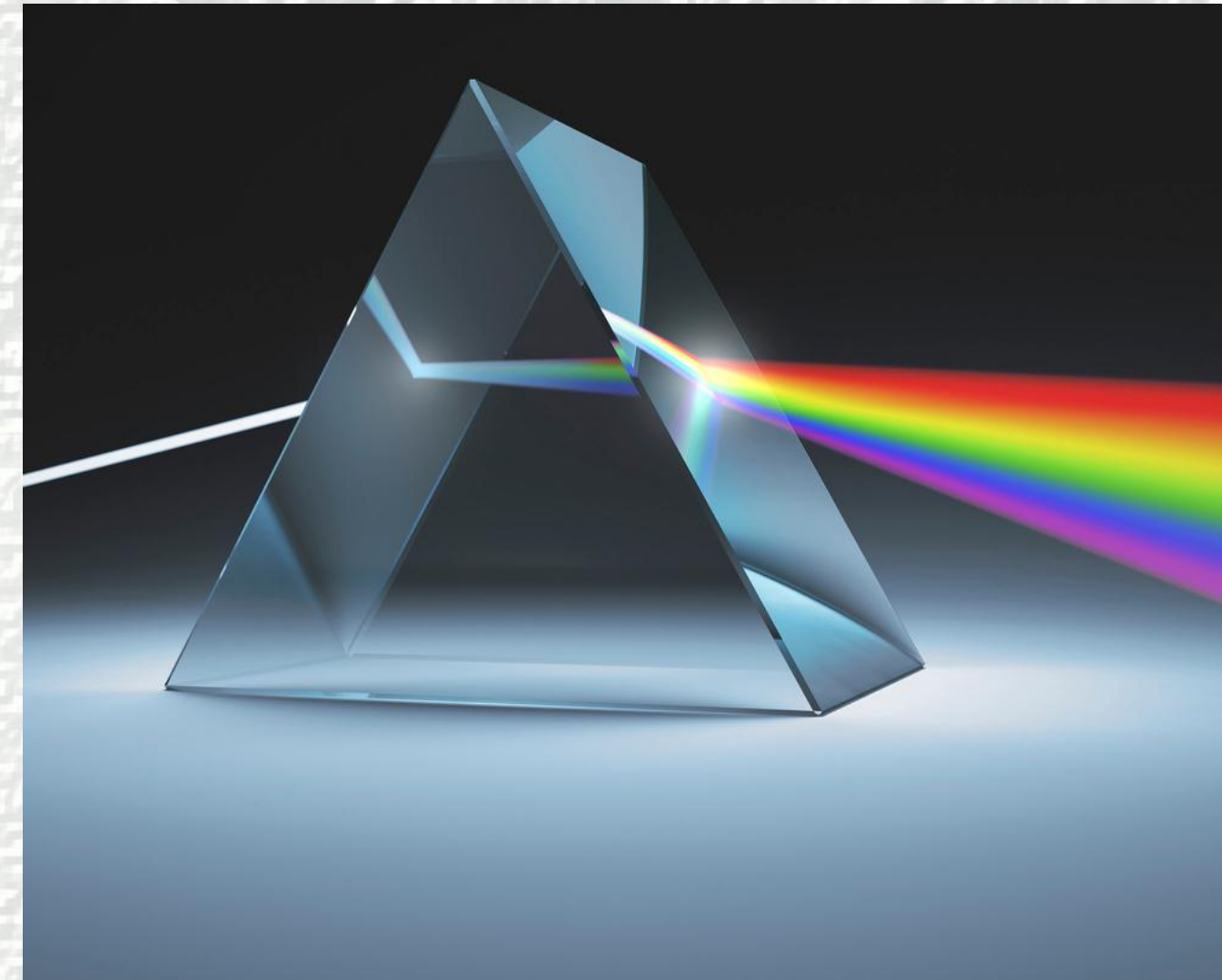
- Refraction is the bending of light waves when it passes from one medium into another (e.g., from air into water).
- Why it happens: The light bends because it changes speed abruptly when hitting the new material.
- Real-World Example: A spoon inside a glass of water looks broken or bent at the surface line.



« PROPERTIES OF LIGHT WAVES

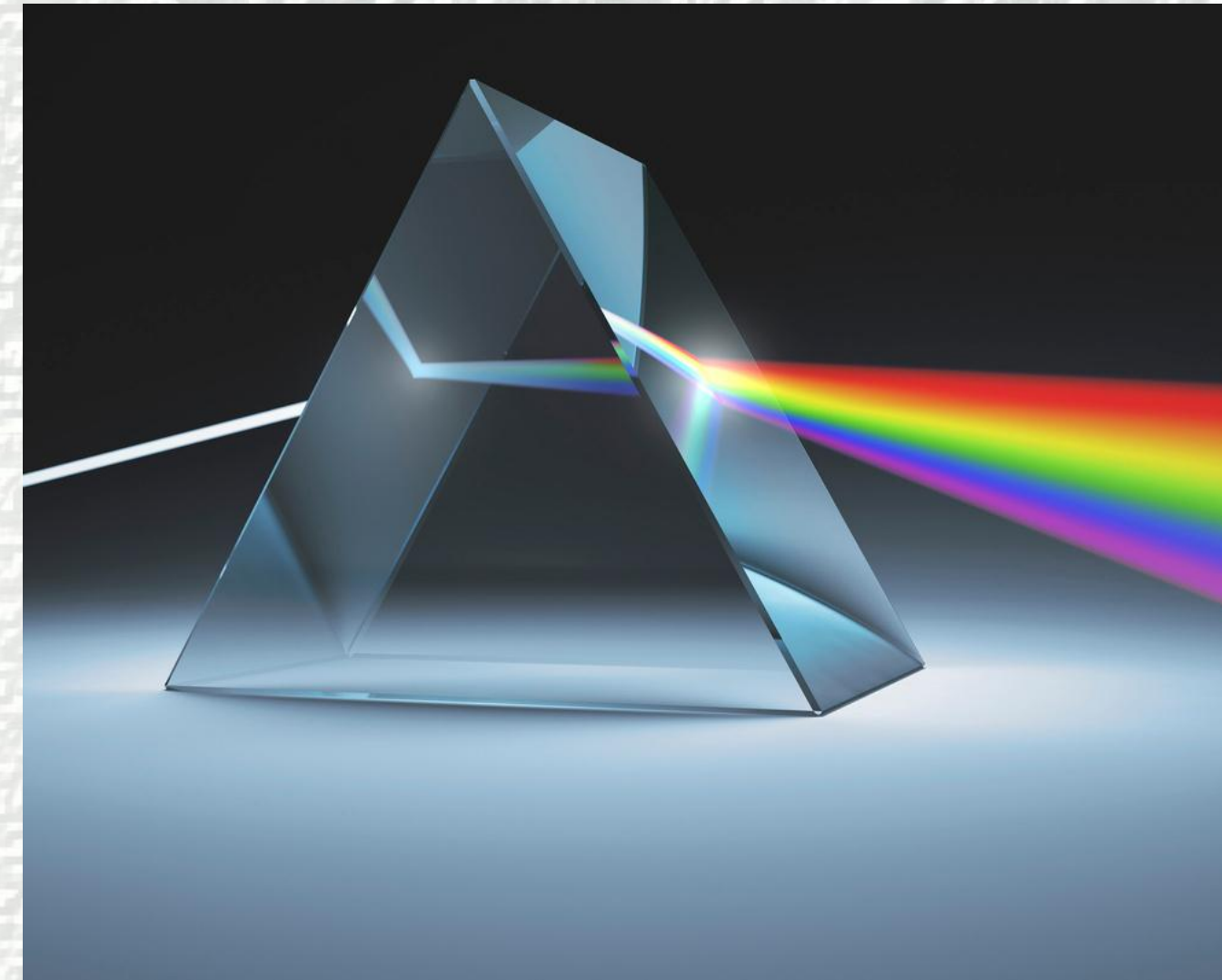
6) DISPERSION

- Dispersion is a special type of refraction. It occurs when white light passes through a prism and splits into its full rainbow of component colors (ROYGBIV).



« PROPERTIES OF LIGHT WAVES

- **White light is actually a mix of all colors. Because different wavelengths (colors) slow down at slightly different rates inside glass, violet light bends the most, and red light bends the least, separating them completely.**



« PROPERTIES OF LIGHT WAVES

DISPERSION

- **Real-World Example: Rainbows forming in the sky after rain when water droplets act as tiny natural prisms.**



« PROPERTIES OF LIGHT WAVES

7) DIFFRACTION

- Diffraction is the bending and spreading of light waves when they pass through a tiny opening (slit) or edge past a sharp obstacle.
- Instead of just casting a perfectly sharp shadow, light waves curve slightly around corners, fanning out like water ripples squeezing through a narrow gap.



« PROPERTIES OF LIGHT WAVES

DIFFRACTION

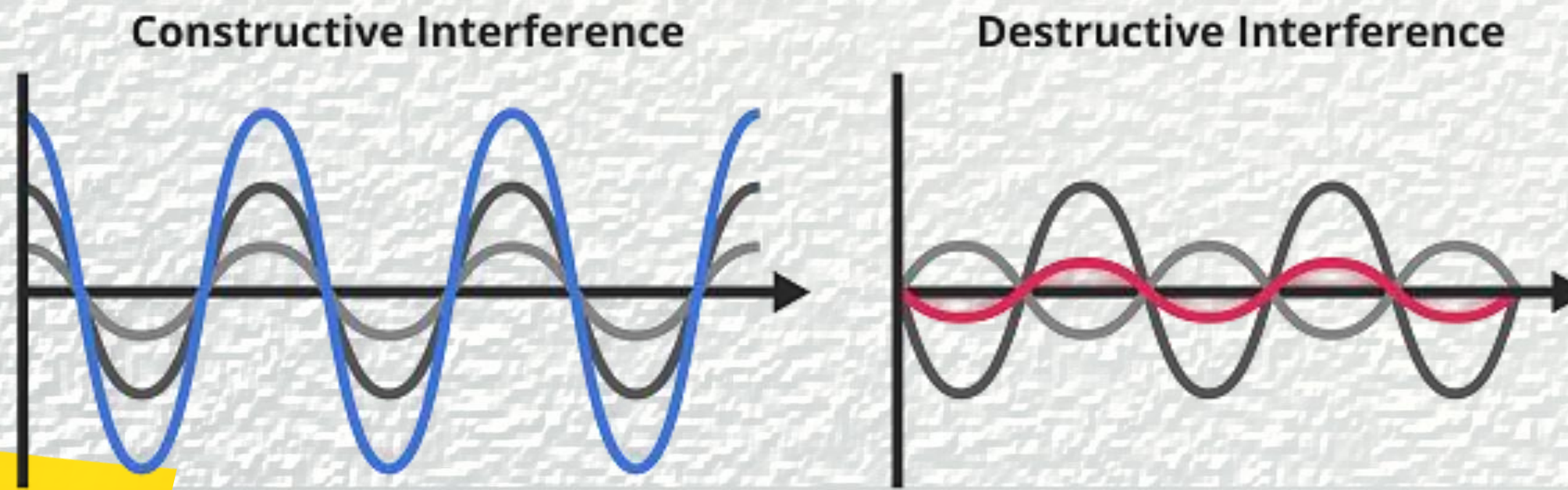
- **Real-World Example:**
The fuzzy edges of shadows, or the rainbow patterns you see looking at the data side of a CD or DVD.



« PROPERTIES OF LIGHT WAVES

8) INTERFERENCE

- Interference happens when two or more light waves meet and overlap with each other, combining to create a new wave pattern.
- Two Types:
 - a. Constructive Interference:** Crest meets crest. The waves reinforce each other, making the light brighter.
 - b. Destructive Interference:** Crest meets trough. The waves cancel each other out, creating darkness/shadow.



« PROPERTIES OF LIGHT WAVES

INTERFERENCE

- **Real-World Example: The swirling, swirling rainbow colors visible on the surface of a soap bubble or an oil slick on wet pavement.**



« PROPERTIES OF LIGHT WAVES

9) POLARIZATION

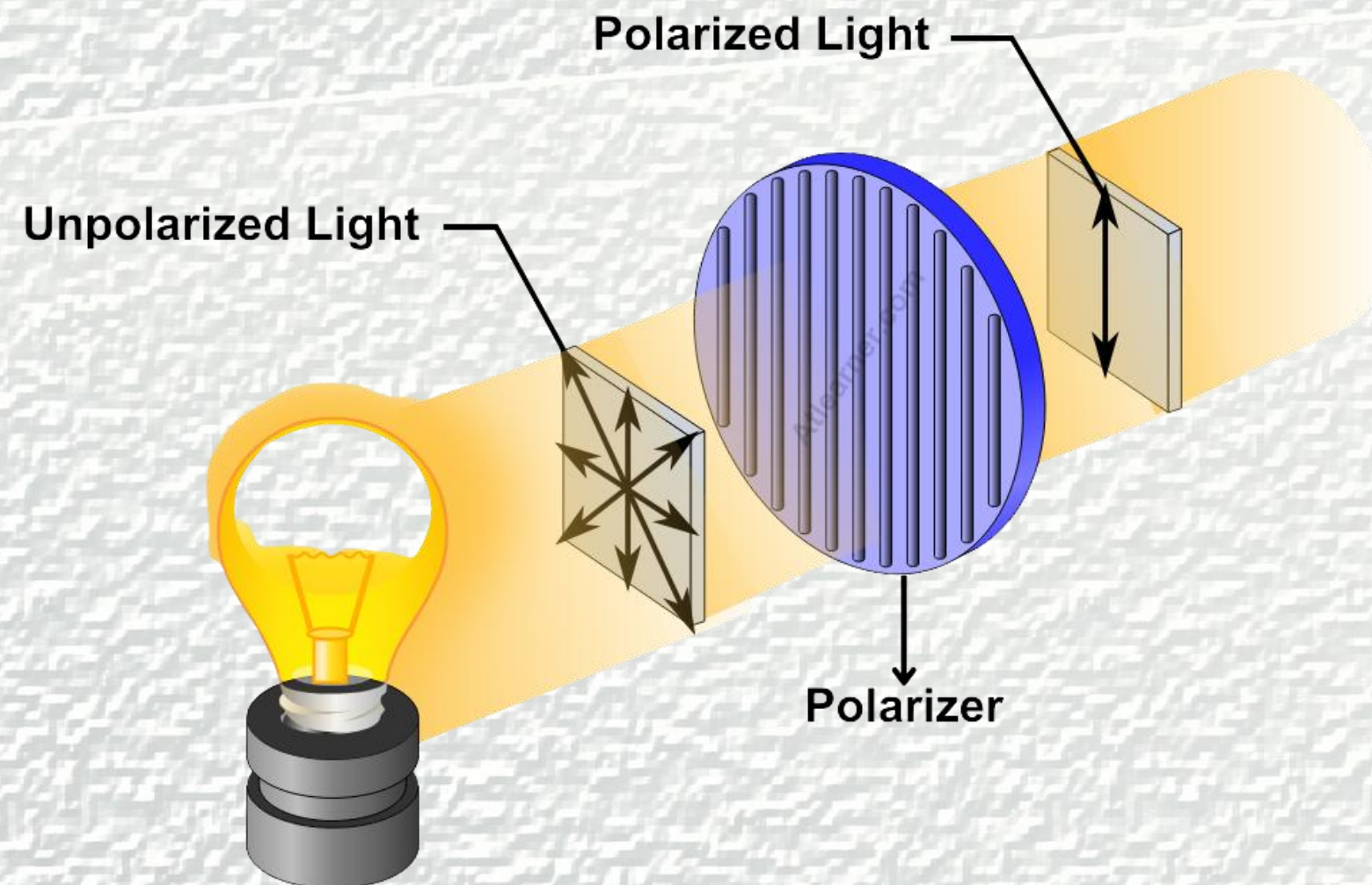
- Normal light is unpolarized, meaning its transverse waves vibrate in all possible up-and-down and side-to-side directions at once. Polarization is the process of filtering light so that it vibrates in only one single plane/direction.



« PROPERTIES OF LIGHT WAVES

POLARIZATION

- **How it works: A polarizing filter acts like a picket fence. It only allows waves vibrating parallel to the slots to pass through, blocking out the rest.**



« PROPERTIES OF LIGHT WAVES

POLARIZATION

- **Real-World Example: Polarized sunglasses that block harsh horizontal glare reflecting off car hoods or water surfaces; 3D glasses at movie theaters.**



« PROPERTIES OF LIGHT WAVES

- **Wavelength:** The physical distance from one wave peak (crest) to the very next peak.
- **Frequency:** The number of wave cycles that pass a point in one second, which determines the light's color and energy.
- **Speed:** Light travels at its ultimate cosmic speed limit (300,000,000 m/s) in a vacuum, but slows down when passing through materials like glass or water.



« PROPERTIES OF LIGHT WAVES

- **Reflection:** The bouncing back of light rays when they strike a surface, following the rule that the incoming angle equals the outgoing angle.
- **Refraction:** The bending of light waves as they pass from one transparent material into another due to an abrupt change in speed.
- **Dispersion:** The splitting of white light into its full rainbow spectrum because different colors bend at slightly different angles inside a prism.



« PROPERTIES OF LIGHT WAVES

- **Diffraction:** The bending and spreading out of light waves as they squeeze through narrow openings or skim past sharp obstacles.
- **Interference:** The overlapping of multiple light waves that either combine to create brighter light or cancel each other out to create dark spots.
- **Polarization:** The process of filtering light waves so that they vibrate in only one single direction or plane.



« PROPERTIES OF LIGHT WAVES

- **Diffraction:** The bending and spreading out of light waves as they squeeze through narrow openings or skim past sharp obstacles.
- **Interference:** The overlapping of multiple light waves that either combine to create brighter light or cancel each other out to create dark spots.
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INTRODUCTION TO SOUND AND ITS PROPERTIES



LEARNING OBJECTIVES:

- 1. Explain how sound behaves as a mechanical, longitudinal wave and describe why it requires a physical medium to travel.**
- 2. Differentiate between the various properties of sound, such as frequency/pitch, amplitude/loudness, and timbre**



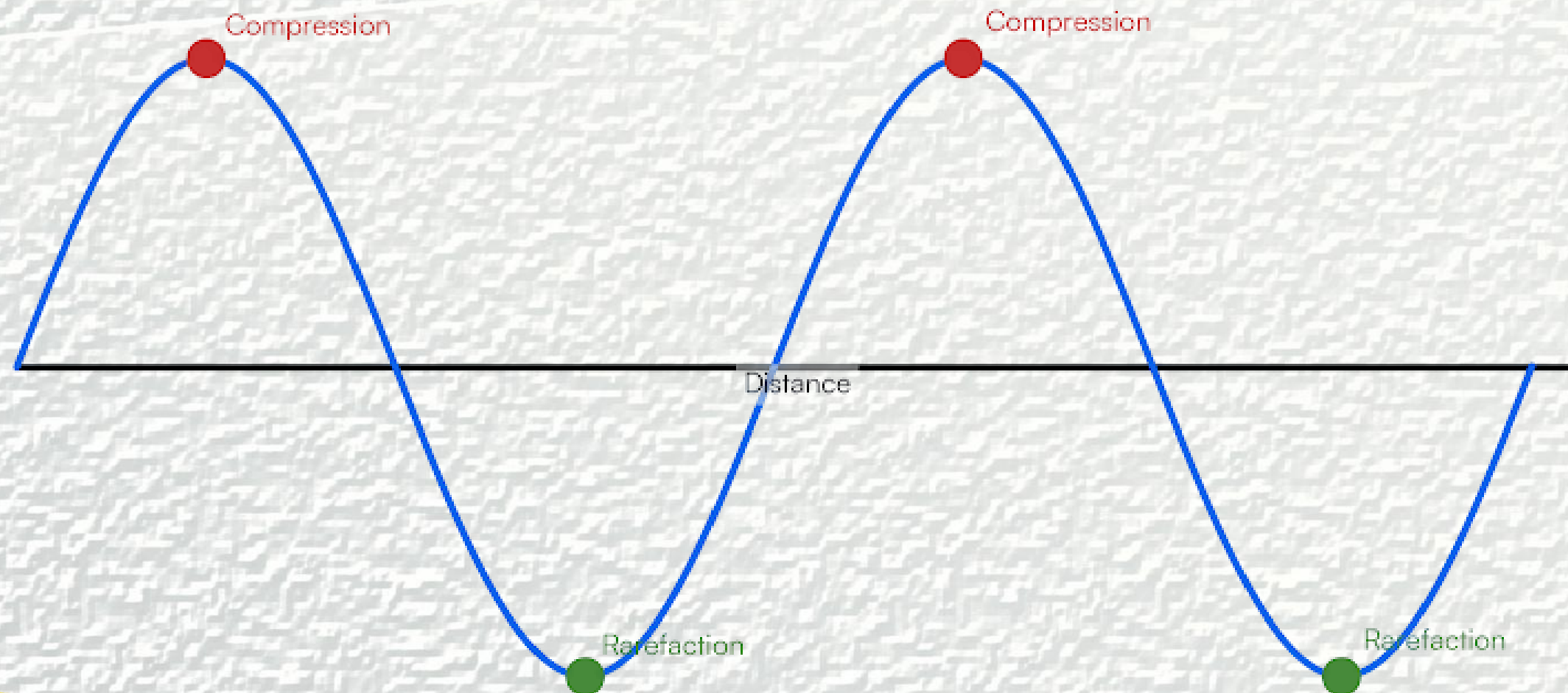
« SOUND IS A MECHANICAL, LONGITUDINAL WAVE

- **Mechanical Wave:** A wave that strictly requires a physical medium (solids, liquids, or gases) to travel. It cannot exist in the empty vacuum of space.
- **Longitudinal Wave:** A wave where particles vibrate back and forth in the same direction that the wave is traveling.



« THE STRUCTURE OF SOUND

- **Compression:** The high-pressure areas where air particles are smashed tightly together.
- **Rarefaction:** The low-pressure areas where air particles are spread far apart.

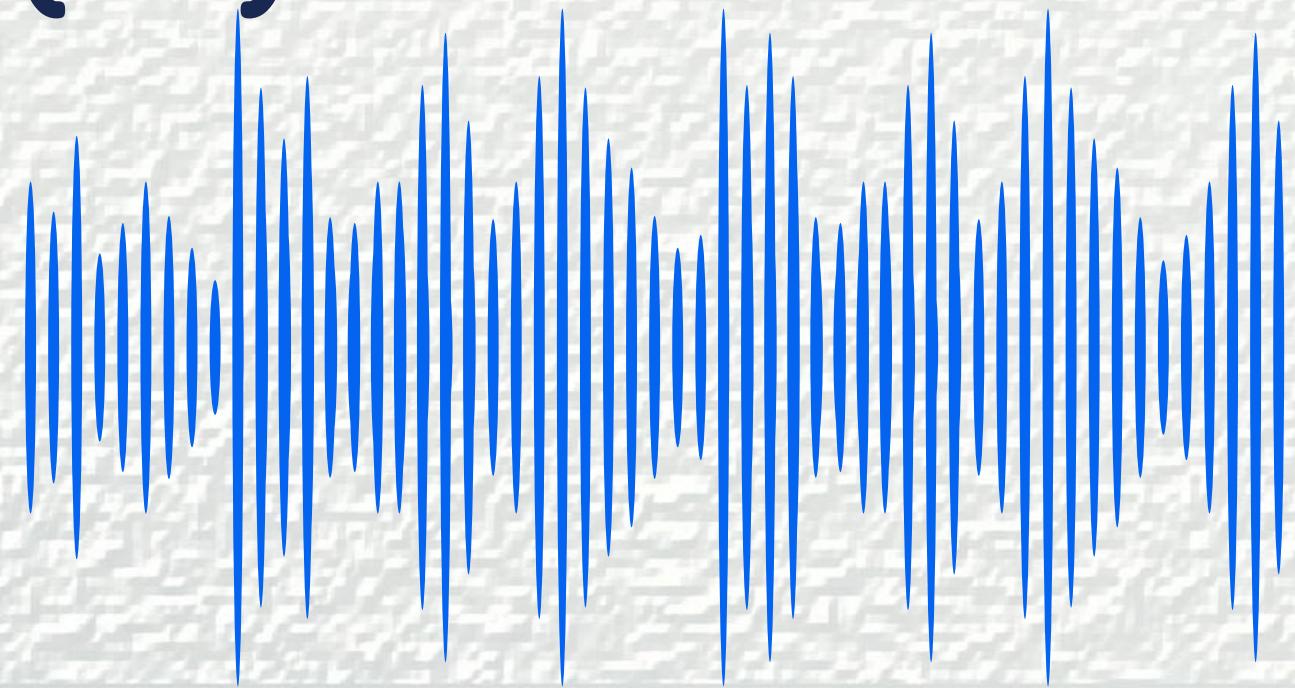


« PROPERTIES OF SOUND

FREQUENCY AND PITCH

Frequency and Pitch are partners—they are directly connected to each other.

- **Frequency:** This is the scientific measurement. It is the number of sound waves that pass a point in one second, measured in Hertz (Hz)





PROPERTIES OF SOUND

FREQUENCY AND PITCH

- **Pitch:** This is what your ears actually perceive.
 - **High Frequency = High Pitch:** The waves are crowded tightly together. Think of a whistling kettle, a mouse squeaking, or a high note on a guitar.
 - **Low Frequency = Low Pitch:** The waves are spaced far apart. Think of a booming bass speaker, a dog growling, or a deep rumble of thunder.



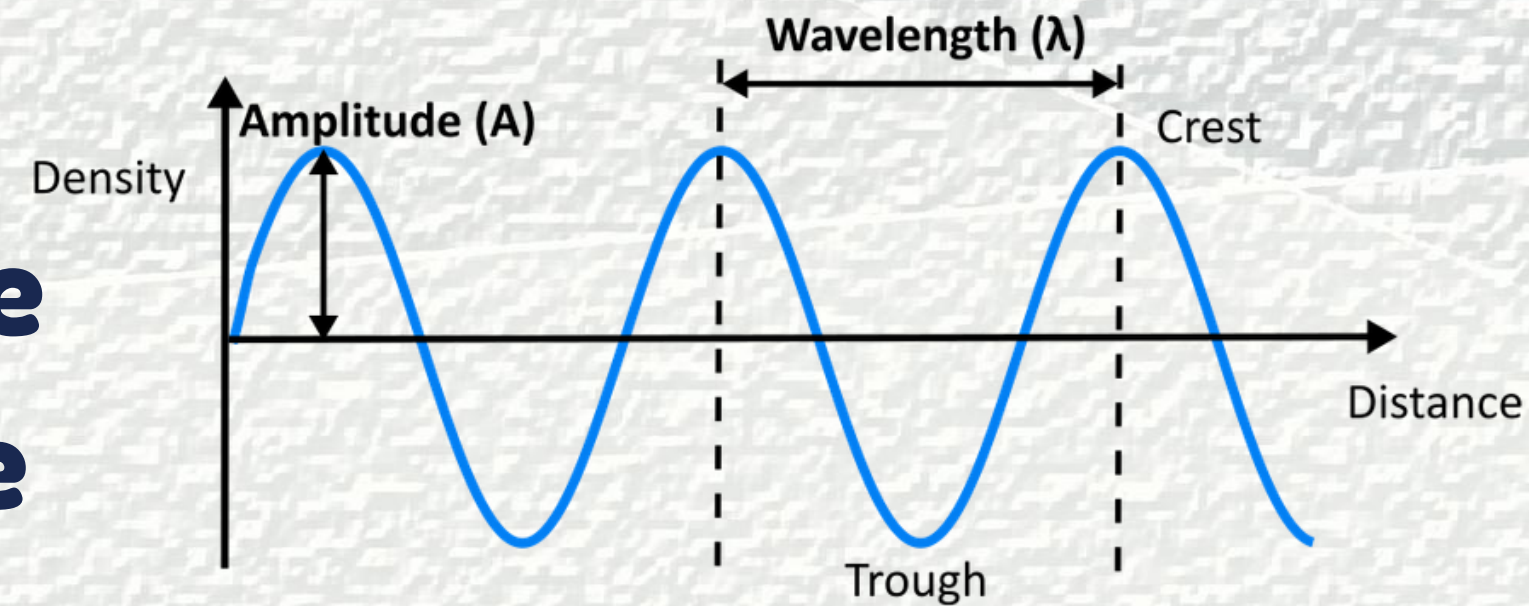
« PROPERTIES OF SOUND

WAVELENGTH

Wavelength is the physical distance between two identical points on back-to-back waves.

- In a sound wave, it is the distance measured from the center of one compression (the squished part) to the center of the very next compression.**

Wavelength of a Sound Wave

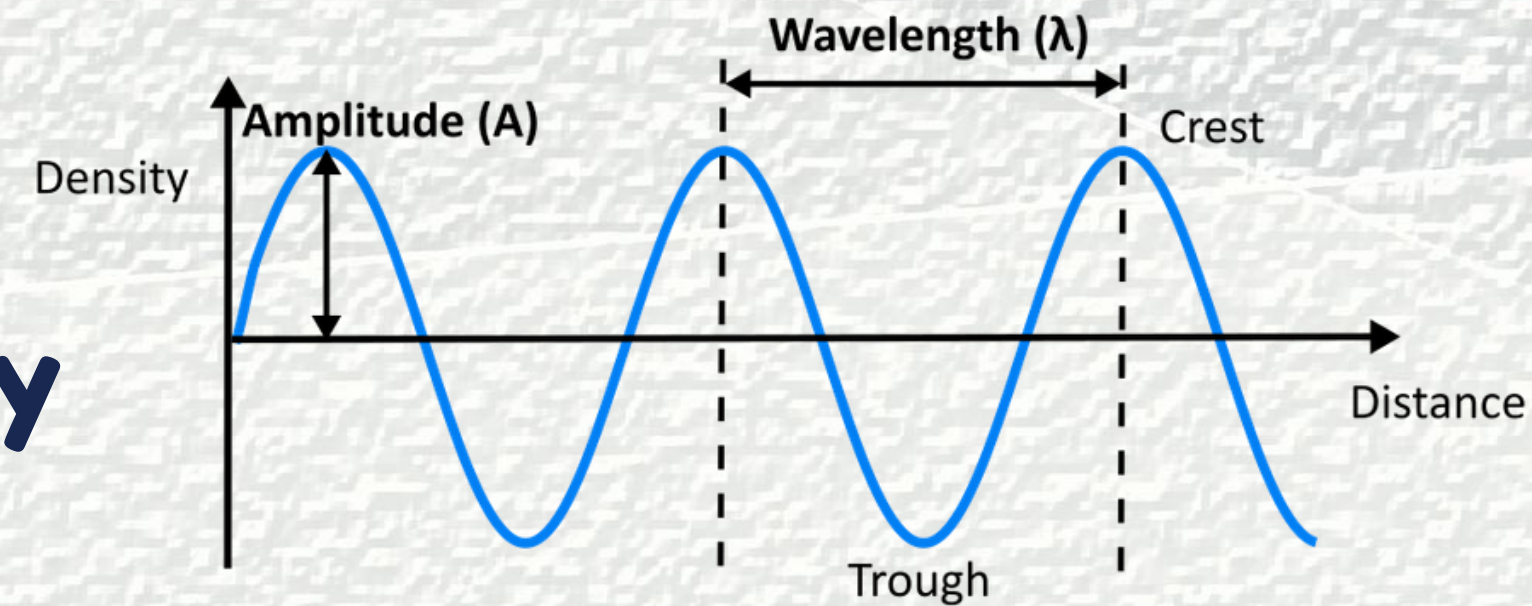


« PROPERTIES OF SOUND

WAVELENGTH

- **The Rule: Wavelength and Frequency have an opposite relationship.**
 - If a sound has a high frequency (high pitch), its wavelength is short.
 - If a sound has a low frequency (low pitch), its wavelength is long.

Wavelength of a Sound Wave

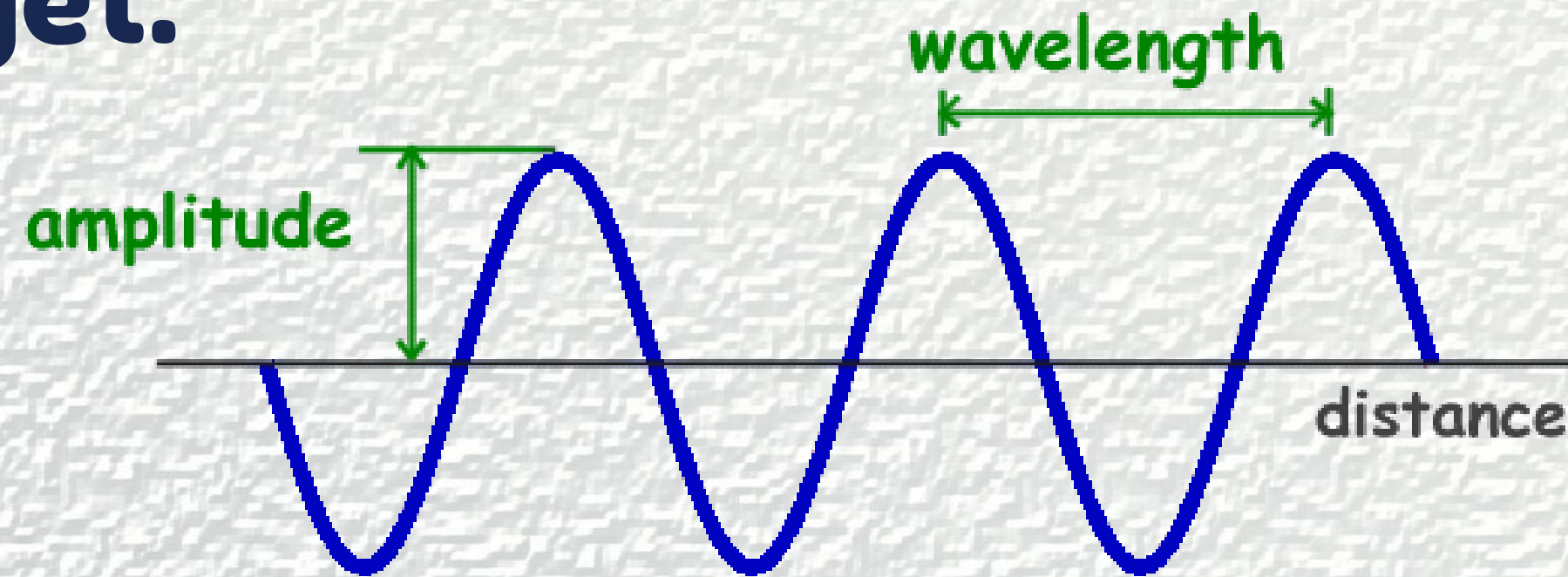


« PROPERTIES OF SOUND

AMPLITUDE AND LOUDNESS

Amplitude and Loudness are partners—they deal with the energy and volume of the sound.

- **Amplitude:** This is the scientific measurement of the wave's power. It is the height of the wave, measuring how tightly compressed the air particles get.



« PROPERTIES OF SOUND

AMPLITUDE AND LOUDNESS

- **Loudness:** This is how loud or soft the sound feels to our ears. It is measured in Decibels (dB).
 - **High Amplitude** = Loud Sound: The wave carries a ton of energy. (Example: A shouting teacher or a motorcycle engine).
 - **Low Amplitude** = Soft Sound: The wave carries very little energy. (Example: A classroom whisper).

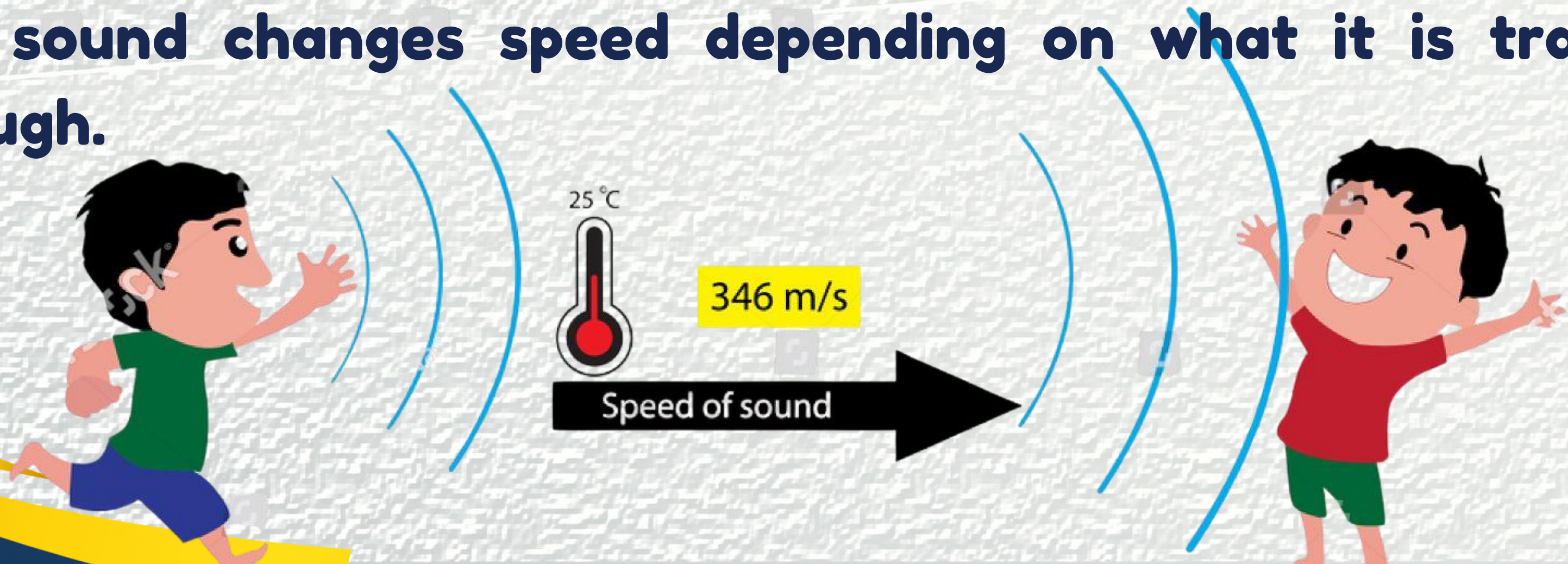


◀ PROPERTIES OF SOUND

SPEED

Speed is how fast the sound energy travels through a space.

- **In normal air, sound travels at about 343 meters per second (m/s).**
- **Sound travels by bumping particles into each other. Because of this, sound changes speed depending on what it is traveling through.**

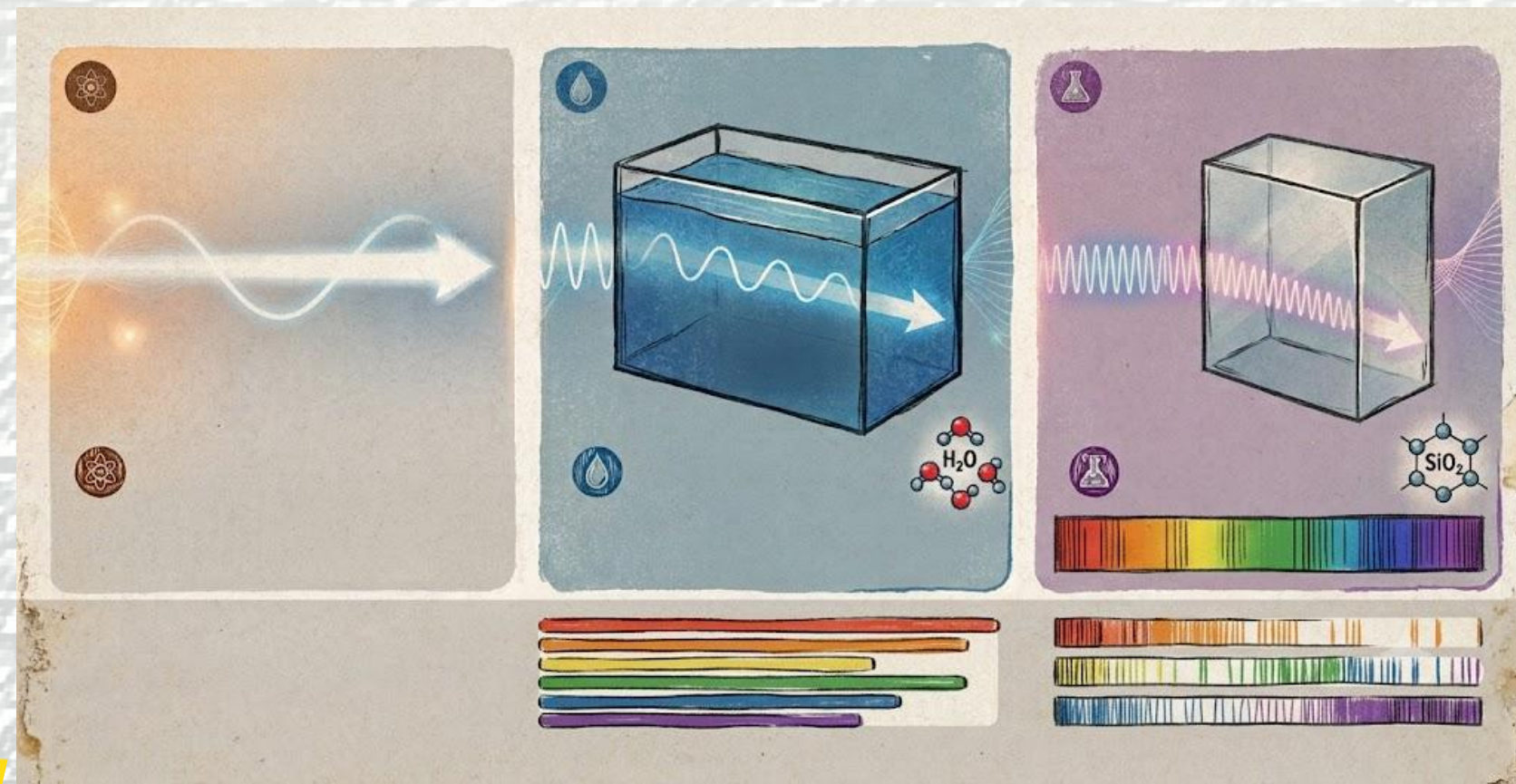




PROPERTIES OF SOUND

SPEED

- **Solids (Fastest):** Particles are tightly packed together, so they pass the vibration incredibly fast. (Putting your ear to a train track lets you hear the train miles away).
- **Liquids (Medium):** Particles are slightly looser.
- **Gases (Slowest):** Particles are far apart, so it takes longer for them to bump into one another.

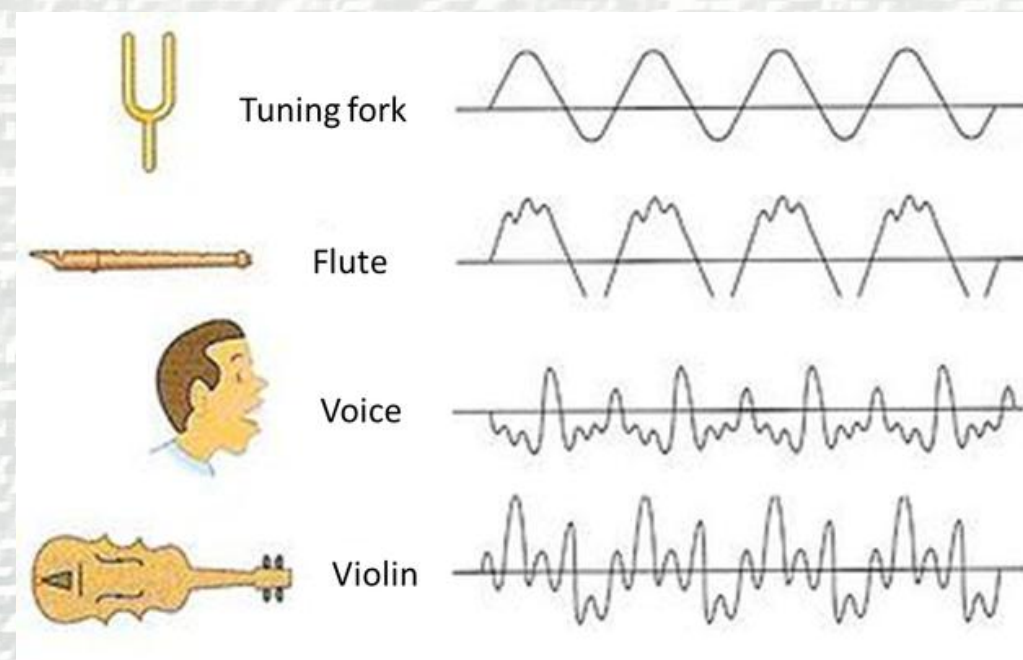


« PROPERTIES OF SOUND

TIMBRE

Pronounced "Tam-ber", this is the quality or "color" of a sound that makes it unique.

- Imagine a piano and a guitar playing the exact same note (same pitch/frequency) at the exact same volume (same amplitude/loudness). You can still easily tell which is the piano and which is the guitar. That difference is timbre.



« PROPERTIES OF SOUND

TIMBRE

- **Real-world sounds are rarely pure, perfect waves. They are a mix of different overlapping wave shapes. Timbre is what helps you distinguish your friend's voice from a stranger's voice, even if they say the exact same word.**



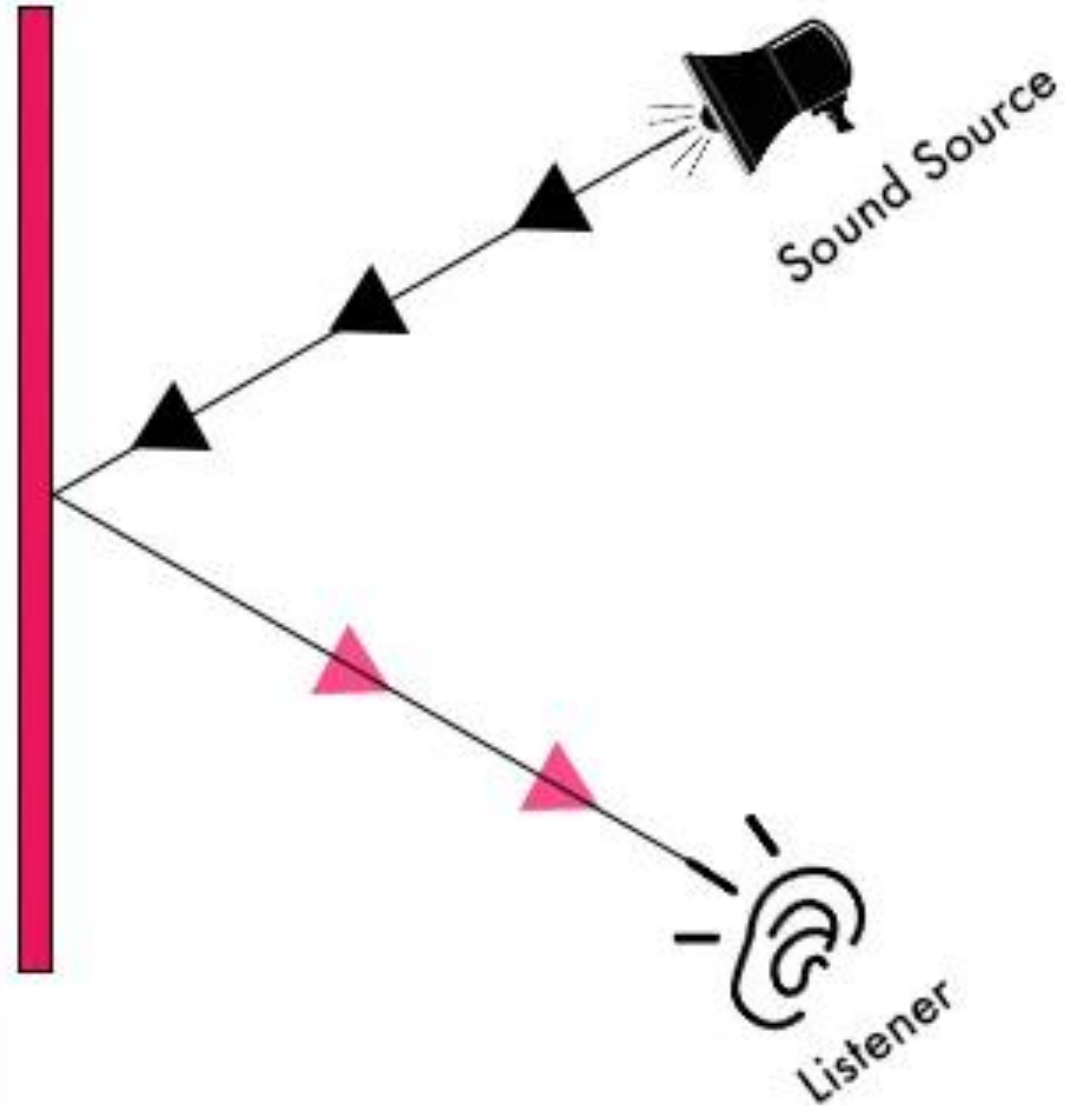
◀ PROPERTIES OF SOUND

REFLECTION, REFRACTION, AND ABSORPTION

- **Reflection (Echoes):** When sound waves hit a hard surface and bounce back. If the surface is far enough away, you hear a distinct echo.
- **Refraction (Bending):** Sound waves bend when they pass through air layers of different temperatures because sound travels faster in warm air than in cold air. (This is why sound travels incredibly far over cold lake water at night).
- **Absorption:** Soft, porous materials (like foam, curtains, or carpets) swallow sound energy and convert it into tiny amounts of heat instead of bouncing it back, deadening the sound in a room.

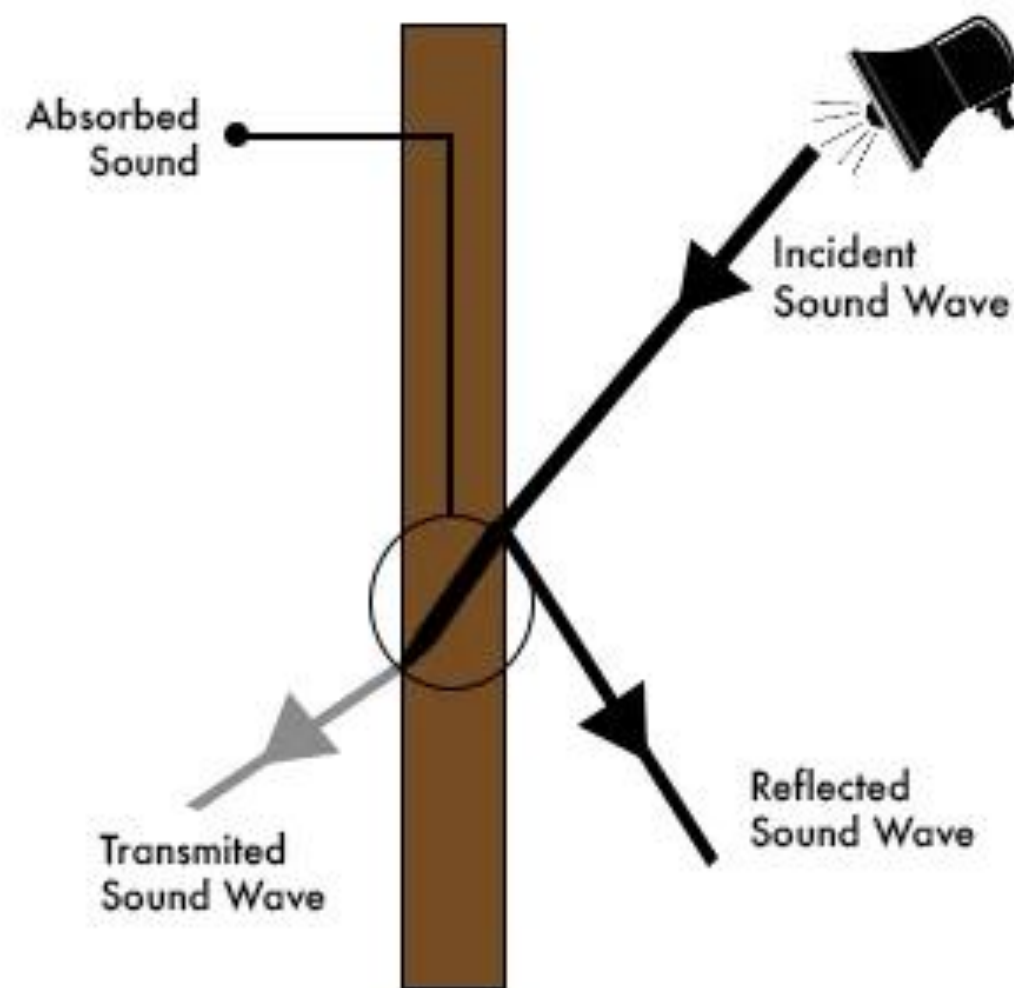


« PROPERTIES OF SOUND



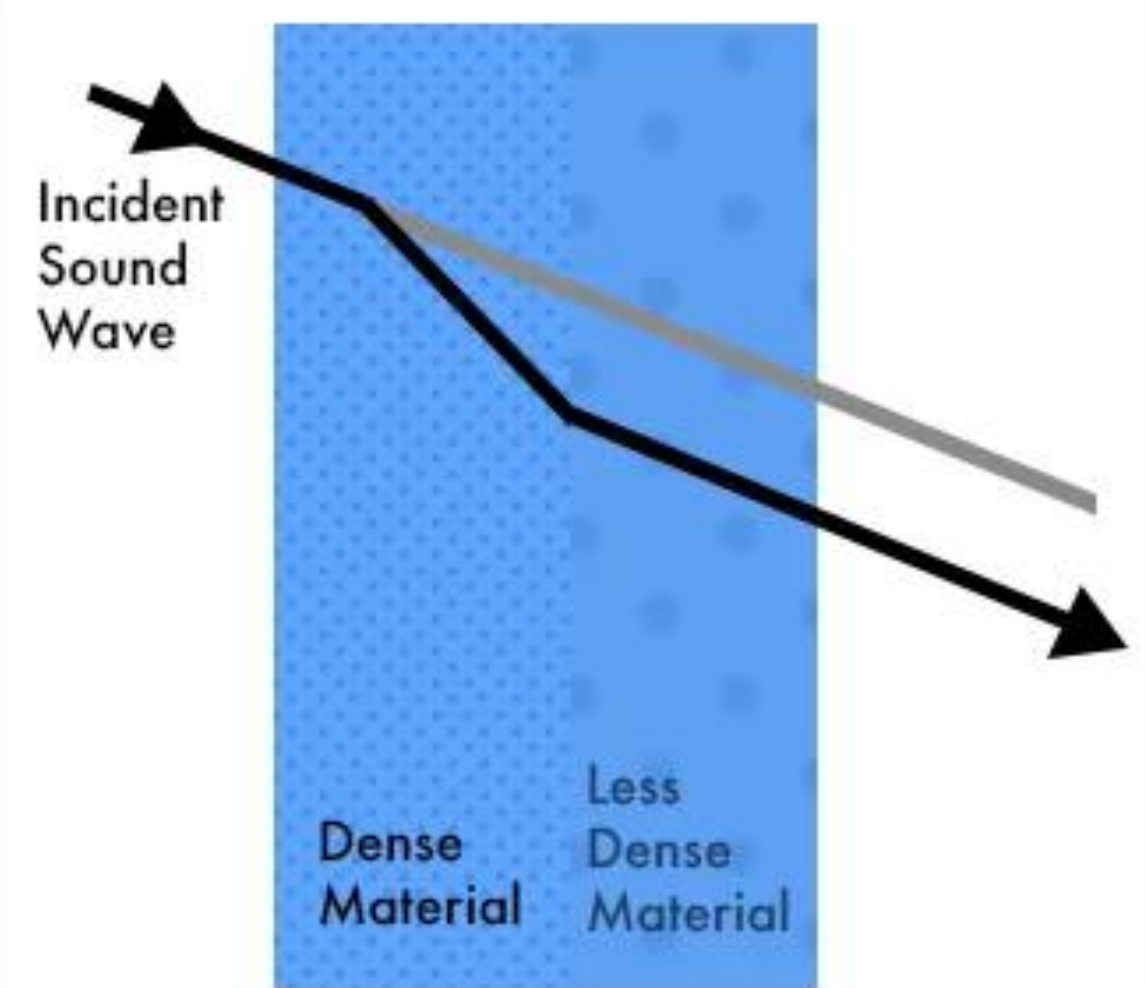
Reflection

When sound that comes in contact with a plain even surface, it bounces back into the environment at exactly the same angle at which it had arrived. Also known as the angle of incidence.



Absorption

Sometimes sound strikes certain materials that absorb a portion of the energy & reflect off the other.



Refraction

When sound passes through a medium of one density into a medium of a different density, the direction of propagation changes, known to have refracted the sound.

◀ REMEMBER

- **Sound is a Physical Collision:** It is a mechanical, longitudinal wave that cannot exist without matter. It relies entirely on a medium (solids, liquids, or gases) to travel. No air means absolute silence!
- **Pitch is Vibe Speed (Frequency):** How fast an object vibrates determines its Frequency (f). High-frequency vibrations pack wave cycles close together (short wavelength) and create a high pitch. Slow vibrations stretch waves out (long wavelength) and create a low pitch.



◀ REMEMBER

- **Loudness is Wave Height (Amplitude):** How hard you push the air determines the wave's Amplitude. Yelling doesn't make sound travel faster across the room—it just makes the wave physically taller, carrying more energy to our ears.
- **Every Sound Has a Unique Fingerprint (Timbre):** Even if two sounds have the identical pitch and volume, their wave textures look different because of the physical shape of what created them. This is why we can instantly tell our classmates' voices apart.





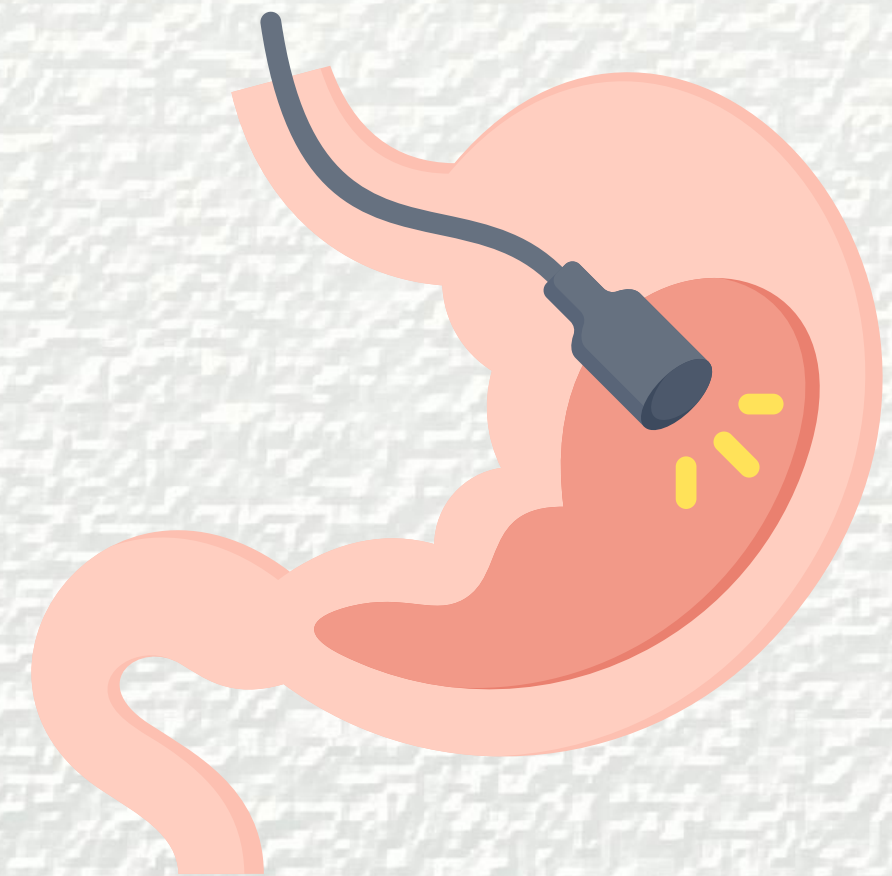
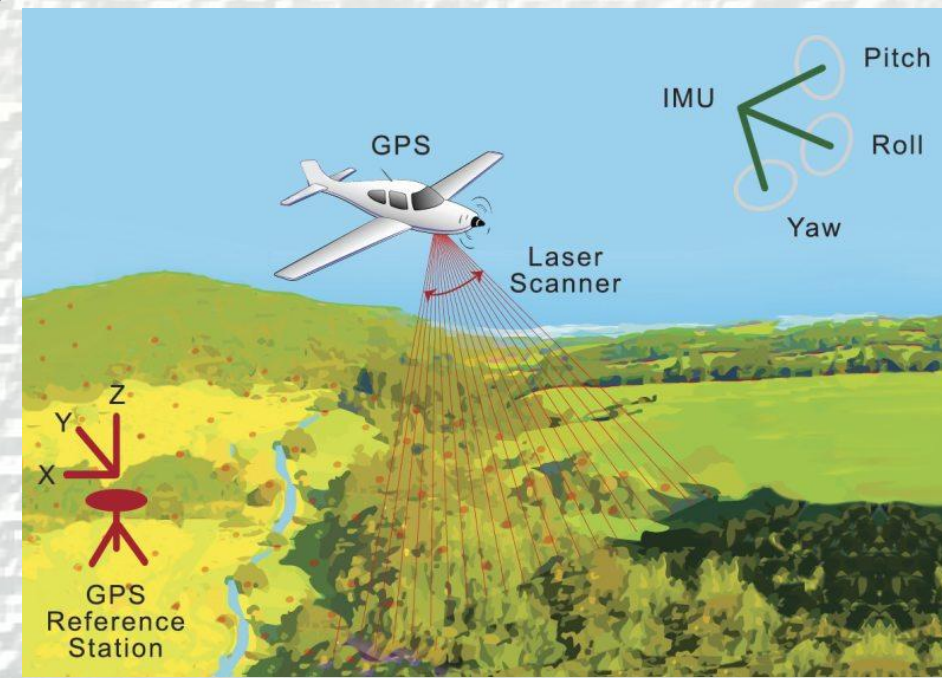
LEARNING OBJECTIVES:

- **Explain how the properties of light and sound waves are engineered to function inside specific technologies, including fiber optics , LIDAR , endoscopes , LASIK , and 3D glasses.**
- **Categorize real-world scenarios under their primary domain pillars: Communication , Navigation , Medicine , or Entertainment.**



« QUESTIONS:

Even though these five technologies look completely different, what is the single secret 'superpower' they all share, and how do they use it to control information?



◀◀ WHAT IS LIGHT?

- **Light** is a form of energy that behaves as an electromagnetic wave.
- Unlike sound waves—which are mechanical and must have physical air or water molecules to bump into—light is self-sustaining. It does not require a medium. Because of this, light can travel through the completely empty vacuum of outer space, allowing sunlight and starlight to reach Earth.



« 4 CORE PROPERTIES OF LIGHT

Speed

- Light is an electromagnetic wave that does not need a medium; it can travel through a vacuum. In empty space, it moves at the cosmic speed limit:

$$c \approx 300,000,000 \text{ m/s}$$

- It slows down predictably when passing through matter like water or glass.



◀ 4 CORE PROPERTIES OF LIGHT

Reflection

When light hits a smooth surface, it bounces back. It follows the Law of Reflection: the incoming angle always equals the bouncing angle.

Refraction

When light passes from one material into another (like air into water), it changes speed. This change in speed causes the light ray to bend at the boundary.



« 4 CORE PROPERTIES OF LIGHT

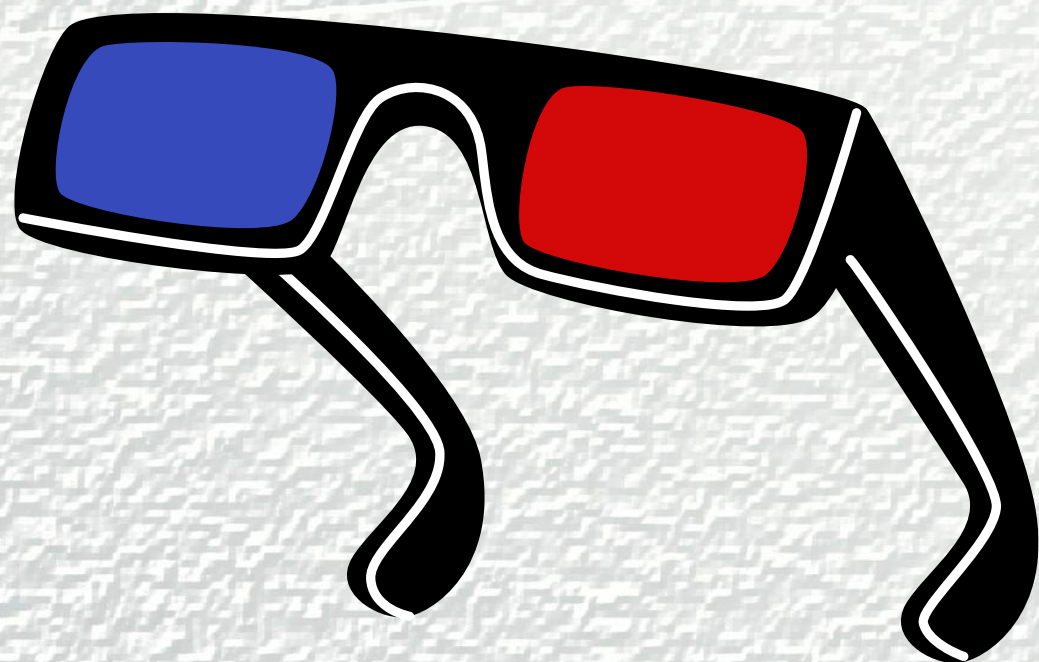
Polarization

Natural light waves vibrate in all directions. Polarization uses a special filter to block unwanted angles, restricting the light waves to vibrate in only one single plane (e.g., only vertically).



FROM PHYSICS TO PHYSICAL REALITY

How do we control the properties of light?



« 4 APPLICATIONS OF LIGHT

 **Communications Technology:** Twisting light around corners to carry the internet.

 **Navigation:** Bouncing laser light to map the physical world in total darkness.

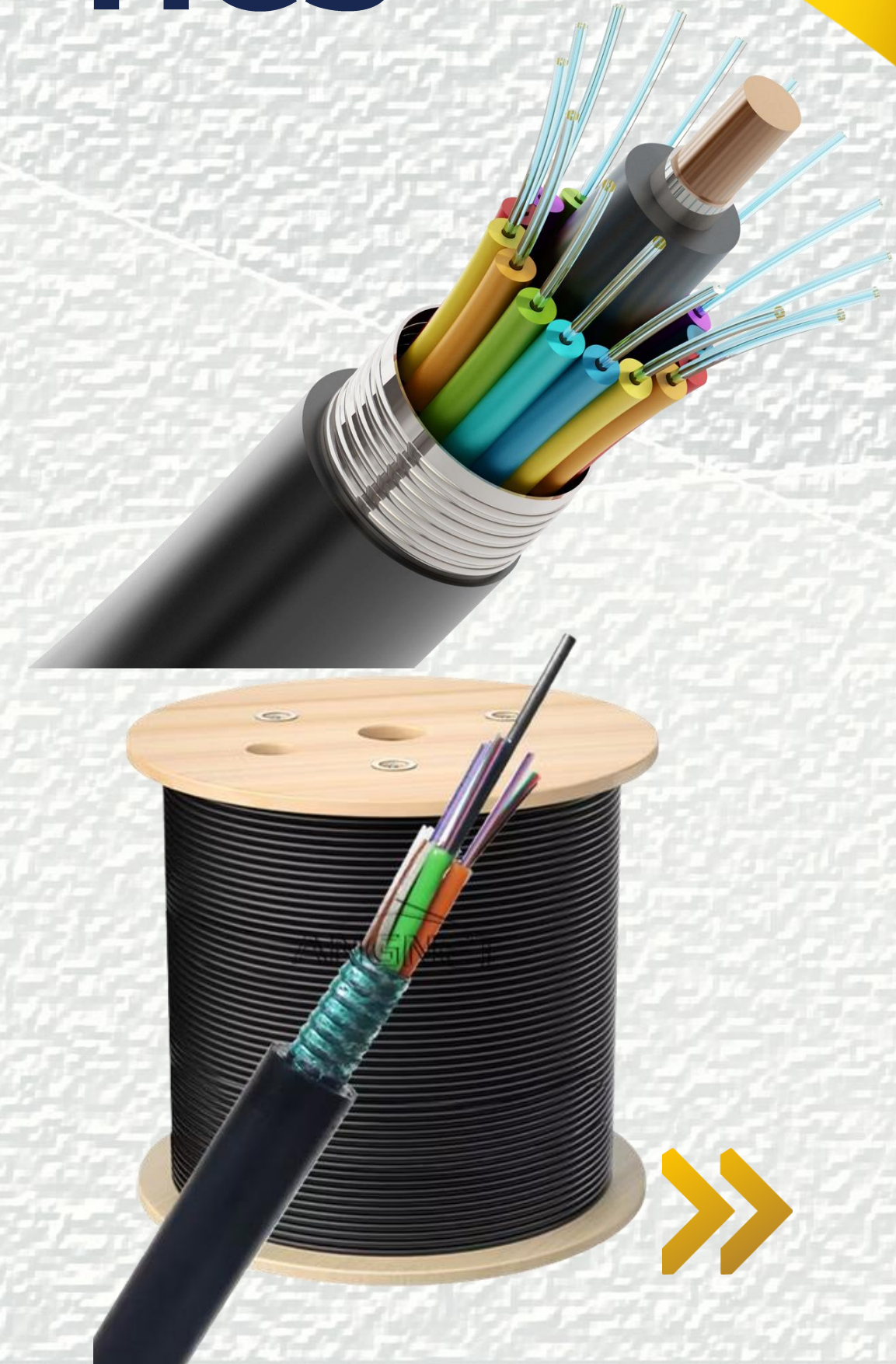
 **Medicine:** Using focused light frequencies to heal the human body without steel tools.

 **Entertainment:** Filtering wave directions to create the illusion of three dimensions.



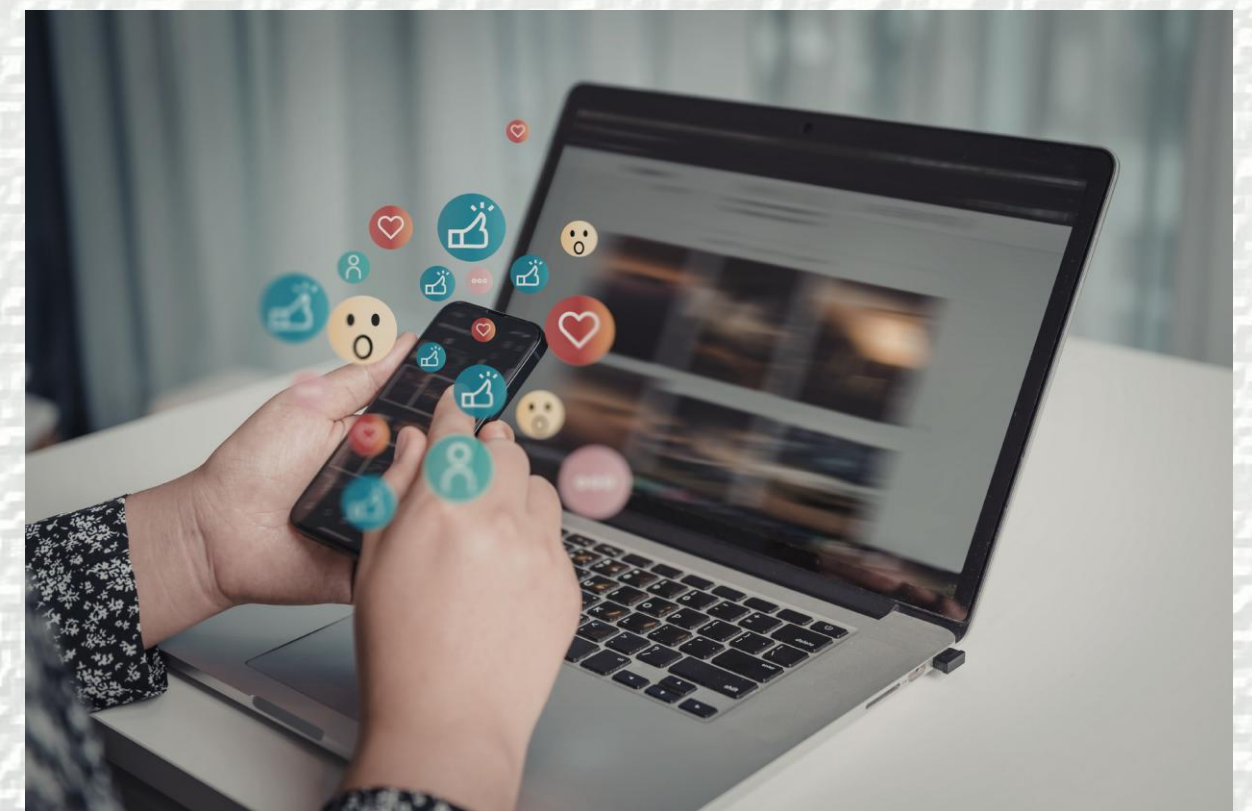
« COMMUNICATIONS TECHNOLOGY (FIBER OPTICS)

- **The Application:** High-speed internet, cable television, and global telecommunication data travel via Fiber-Optic Cables.
- **How It Works:** Massive data streams are converted into binary codes of light flashes. These flashes travel through microscopic glass fibers.



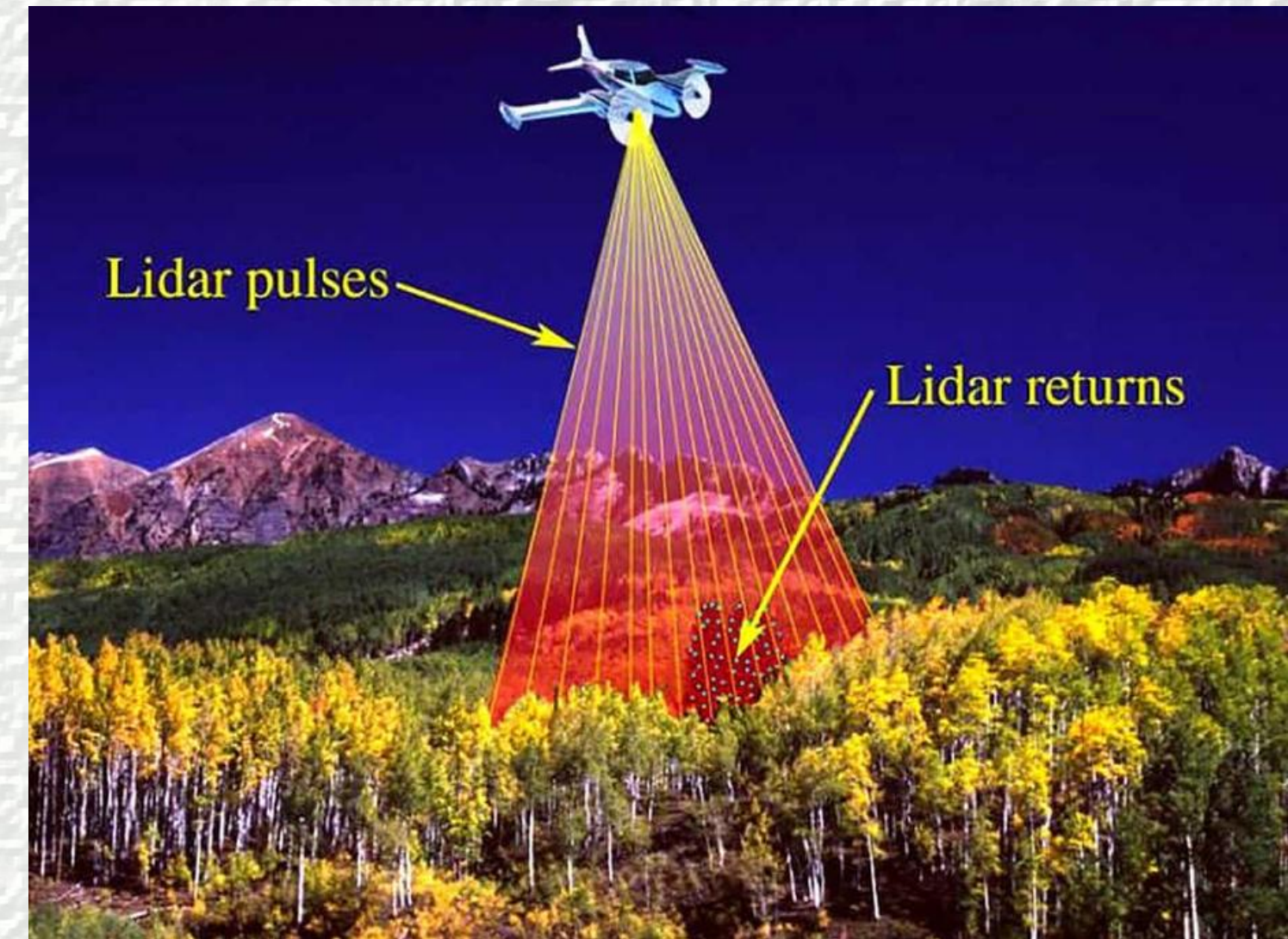
« COMMUNICATIONS TECHNOLOGY (FIBER OPTICS)

- **The Light Property:** Refraction and Reflection (Total Internal Reflection). The glass core is wrapped in a material that forces the light to continually bounce off the internal walls without escaping, allowing data to zigzag across oceans at near light speed.



« NAVIGATION (LIDAR TECHNOLOGY)

- **The Application:** LIDAR (Light Detection and Ranging) is used by self-driving autonomous vehicles, topographical drones, and military navigation mapping.
- **How It Works:** A sensor fires millions of invisible, infrared laser pulses every second in a 360-degree circle.



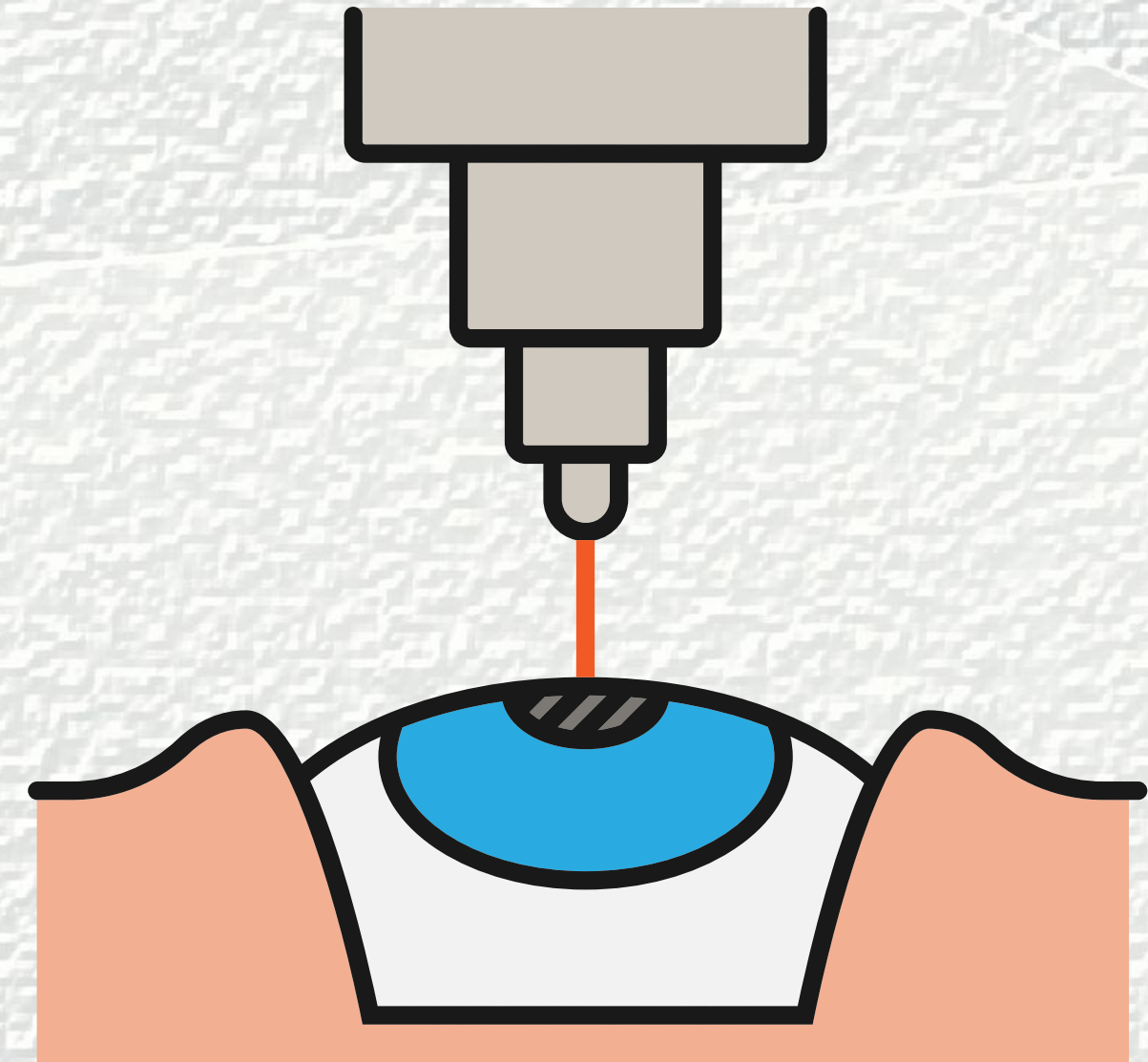
◀ NAVIGATION (LIDAR TECHNOLOGY)

- **The Light Property:** Speed and Reflection. The laser pulses bounce off objects (cars, pedestrians, trees) and return to the sensor. Because the speed of light is a known constant, a computer calculates exactly how far away objects are based on how many nanoseconds the reflection took to return, creating a live 3D map.



◀ MEDICINE

- **The Application:** Non-invasive surgeries (Endoscopy) and corrective procedures (LASIK eye surgery).



◀ MEDICINE

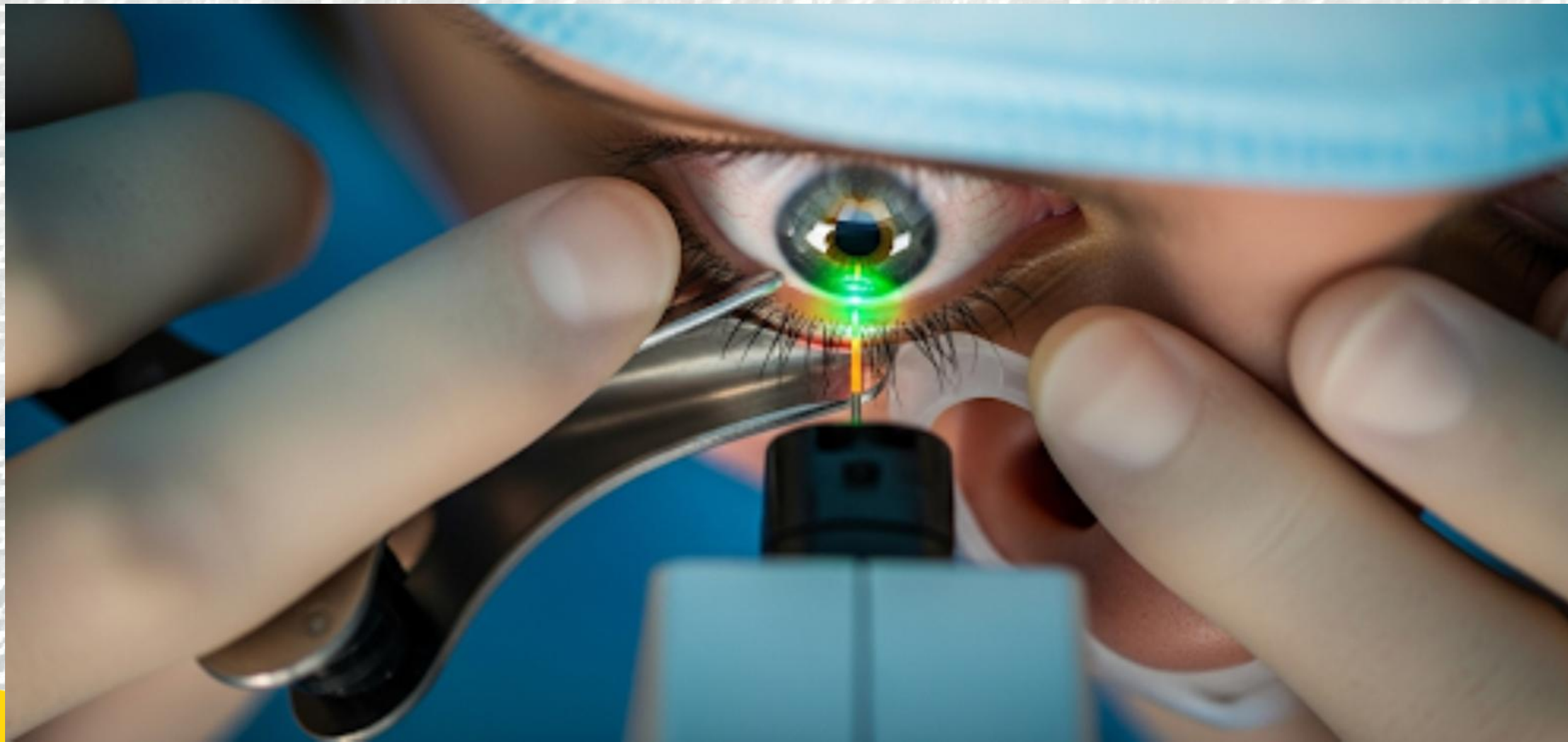
How It Works:

- Endoscopes use two bundles of fiber optic threads. One carries bright light down into a patient's body cavity, and the other carries the reflected visual image back out to a screen.



« MEDICINE

- **LASIK Surgery uses ultra-focused, single-frequency UV laser beams to precisely reshape the cornea of the eye to fix vision problems.**



« MEDICINE

- **The Light Property:** Refraction and Intensity. Doctors manipulate how light curves through lens assemblies to magnify target zones, or intensify its amplitude to burn microscopic tissue paths cleanly without using steel scalpels.



« ENTERTAINMENT

- **The Application:** Passive 3D movies in modern cinemas.
- **How It Works:** The projector displays two distinct perspectives of the movie onto the screen simultaneously. One image is projected with vertically vibrating light; the other is projected with horizontally vibrating light. Without glasses, it looks like a blurry double image.



« ENTERTAINMENT

- **The Light Property:** Polarization. 3D glasses have two different lens filters. The left lens blocks horizontal light, and the right lens blocks vertical light. Each eye is forced to see only one designated perspective, causing your brain to construct a simulated three-dimensional environment.





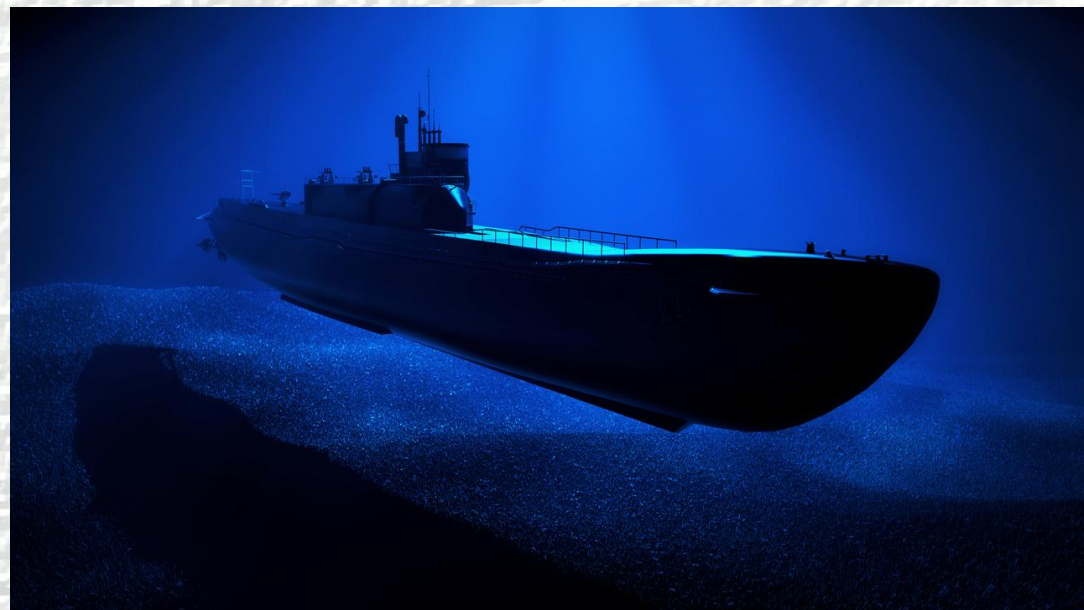
LEARNING OBJECTIVES:

- **Explain how the physical mechanisms of sound waves (such as transducers, constant speed, reflection, and wave interference) are utilized to drive everyday technologies.**
- **Evaluate diverse real-world case studies to determine whether a sound-based technology is functioning as a tool for Communication, Navigation, Medicine, or Entertainment.**



PROPERTIES OF SOUND

AND ITS APPLICATIONS



« COMMUNICATION

- **The Application:** Microphones, Telephones, and Public Address Systems.
- **How it Works:** Sound waves are vibrations in the air.



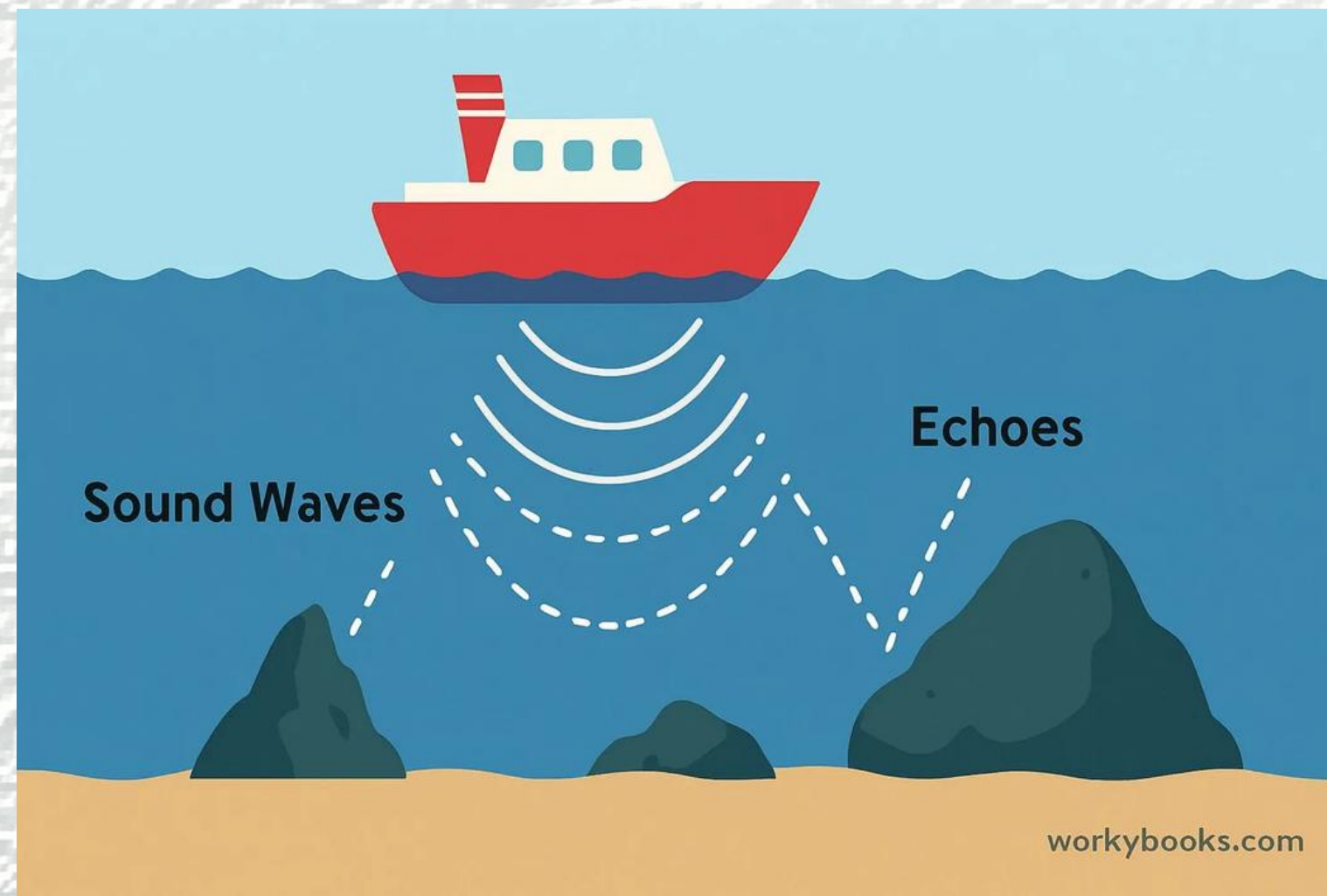
« COMMUNICATION

- **The Physics:** A Transducer (like the diaphragm in your phone) captures the physical vibration of your voice and converts it into an electrical signal. That signal is sent across wires or waves and then "unpacked" back into physical vibrations by a speaker at the other end.



NAVIGATION

- **The Application:** Submarines, Ship depth-finders, and Search & Rescue drones.
- **How it Works:** SONAR (Sound Navigation and Ranging).





Sonar, sound navigation and ranging

« NAVIGATION

- **The Physics (Speed + Reflection):** Just like LIDAR uses light, SONAR sends a "ping" of sound into the water. The sound hits an object and bounces back (Reflection). Because the Speed of sound in water is constant, the computer measures the time it took for the echo to return to calculate the exact distance and shape of the object.

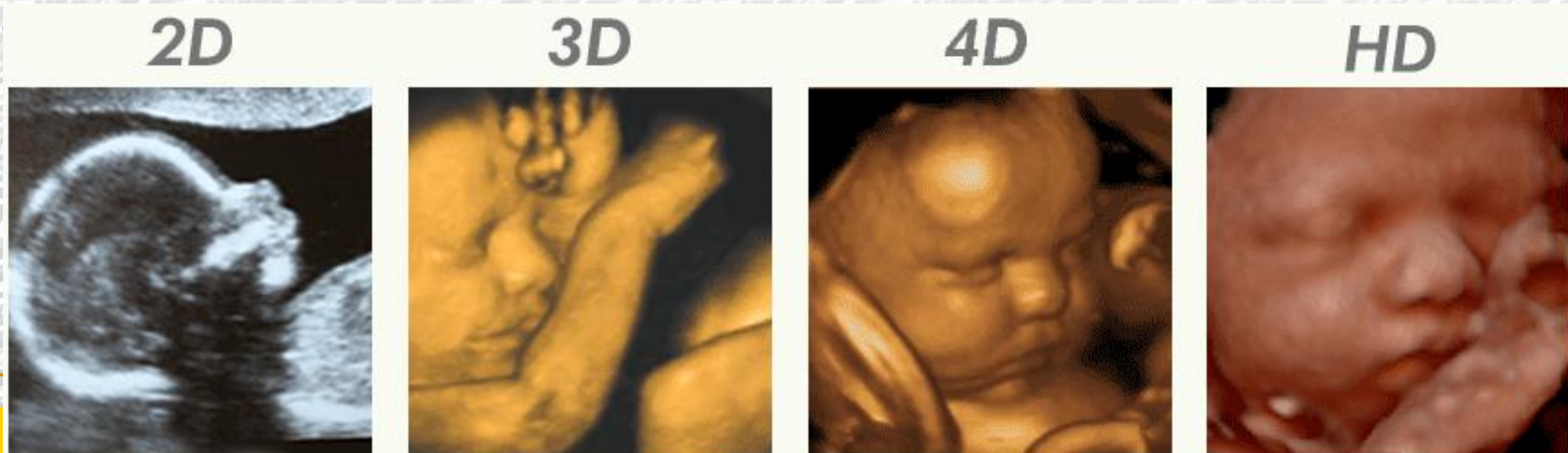
« MEDICINE

- **The Application:** Diagnostic Imaging (Obstetrics, Cardiology).
- **How it Works:** Using high-frequency sound waves that are too high for humans to hear.



« MEDICINE

- **The Physics (Reflection):** A probe shoots ultrasound waves into the body. When the waves hit different densities (bone vs. muscle vs. fluid), they reflect at different strengths. The computer interprets these "sound shadows" to stitch together a live image of an internal organ or a baby



« ENTERTAINMENT

- **The Application:** Noise-Canceling Headphones and Concert Hall Design.
- **How it Works:** Controlling how sound waves interact with each other.



« ENTERTAINMENT

- **The Physics
(Interference):**

In Headphones: A microphone listens to outside noise and creates a "mirror" sound wave (Out-of-phase) that cancels out the vibration before it reaches your ear.



« ENTERTAINMENT

- **The Physics (Interference):**
In Concert Halls: Architects use soft materials to absorb sound (preventing unwanted echoes) or curved surfaces to reflect sound toward the audience.





THANK YOU